

INSTITUTE OF MANAGEMENT SCIENCES, PESHAWAR

Department of Computer Science

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Project Title:

**HR Management System for District Court
Orakzai**

(A Standalone Windows Desktop Application)

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1. Project Identification

Project Name: HR Management System for District Court Orakzai (Standalone Windows Desktop Application)

Client: The Office of the District & Sessions Judge, Orakzai, Khyber Pakhtunkhwa.

2. Client Introduction

The District Court of Orakzai is a vital judicial body in the Khyber Pakhtunkhwa province, responsible for providing legal and administrative services to the local population. The court consists of both **Judicial Officers** (District Judges, Additional District Judges, Civil Judges) and a large **Administrative Staff** (Superintendents, Stenos, Clerks, and Naib Qasids).

Currently, the Human Resource department of the court relies on traditional **Service Books** and manual registers to track the professional lifecycles of its employees. This project aims to digitize these judicial administrative workflows to ensure faster decision-making and higher data integrity.

3. Organization Structure

The judicial hierarchy determines the seniority and administrative rights within the court. The system will primarily be operated by the administrative wing under the supervision of the District & Sessions Judge.

Hierarchy Overview:

1. **District & Sessions Judge (D&SJ):** The final authority over HR and administrative decisions.
2. **Additional District & Sessions Judges (AD&SJs):** High-level judicial officers.
3. **Senior Civil Judges / Civil Judges:** Lower-tier judicial officers.
4. **Superintendent / Clerk of Court:** The primary administrative leads who will act as the **Authorized Users** of this system.
5. **Administrative Staff:** Clerks, Stenos, and Support Staff whose records are being managed.

The **Superintendent** is usually the one who actually uses the computer, but the **District & Sessions Judge** is the one who approves the final printed reports.

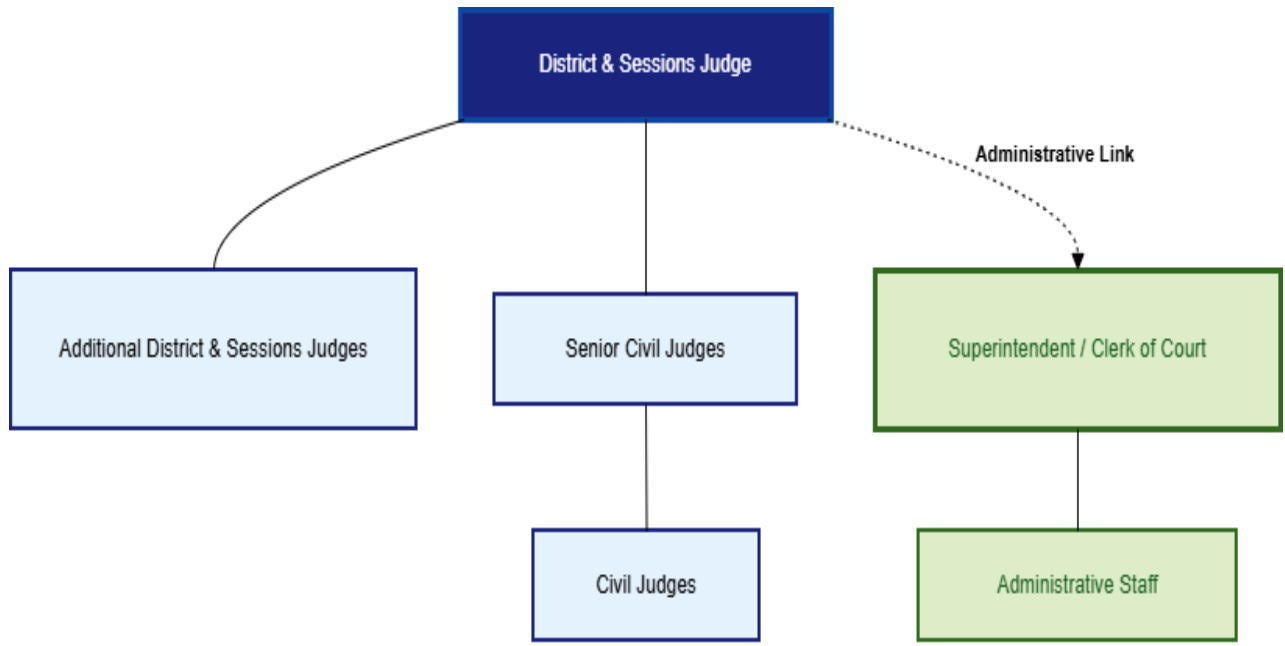


Figure 3.1: Organizational hierarchy of District Court Orakzai, from the District & Sessions Judge down to the administrative staff whose records the system manages.

4. The Process Model

4.1 Requirements for the Process

The development process for a judicial administrative tool must adhere to strict constraints due to the sensitive nature of the data. The requirements for our chosen process include:

- **Predictability:** The ability to map out all 7 use cases and their logic (like the 25-day leave rule) before coding begins.
- **High Verifiability:** Each administrative requirement—especially seniority ranking—must be verified against court rules at every stage.
- **Documentation Rigor:** Since this is for a formal government body, every phase of development must produce clear documentation for future audits.

4.2 Selection Criteria

To choose the most effective model, the following criteria were established:

1. **Scope Stability:** Since the HR rules of the District Court are fixed by law, the model must handle a "Fixed Scope" environment perfectly.
2. **Error Prevention:** The model must prioritize finding logic errors (like incorrect leave balance deductions) early in the cycle.
3. **Resource Efficiency:** As a standalone Windows application, the model should not require the constant, rapid iterations of a web-based "Agile" startup.

4.3 Alternative Models Considered

Before selecting the final process, other common models were analyzed:

Model	Why it was considered	Why it was rejected
Waterfall	Simple, linear, and easy to manage for a single developer.	Too risky; testing only happens at the very end, which could lead to massive rework if the seniority logic is flawed.
Agile (Scrum)	Highly flexible and allows for constant user feedback.	The court's HR rules are not "flexible"—they are fixed. Constant changes to the core logic would be counter-productive for this specific project.
Spiral	Excellent for high-risk projects with many unknowns.	This project has a clear, fixed scope; the high overhead and cost of the Spiral model are unnecessary for a standalone HR tool.

4.4 The Chosen Model: The V-Model

The **V-Model (Verification and Validation Model)** was selected as the most appropriate process for the District Court Orakzai project.

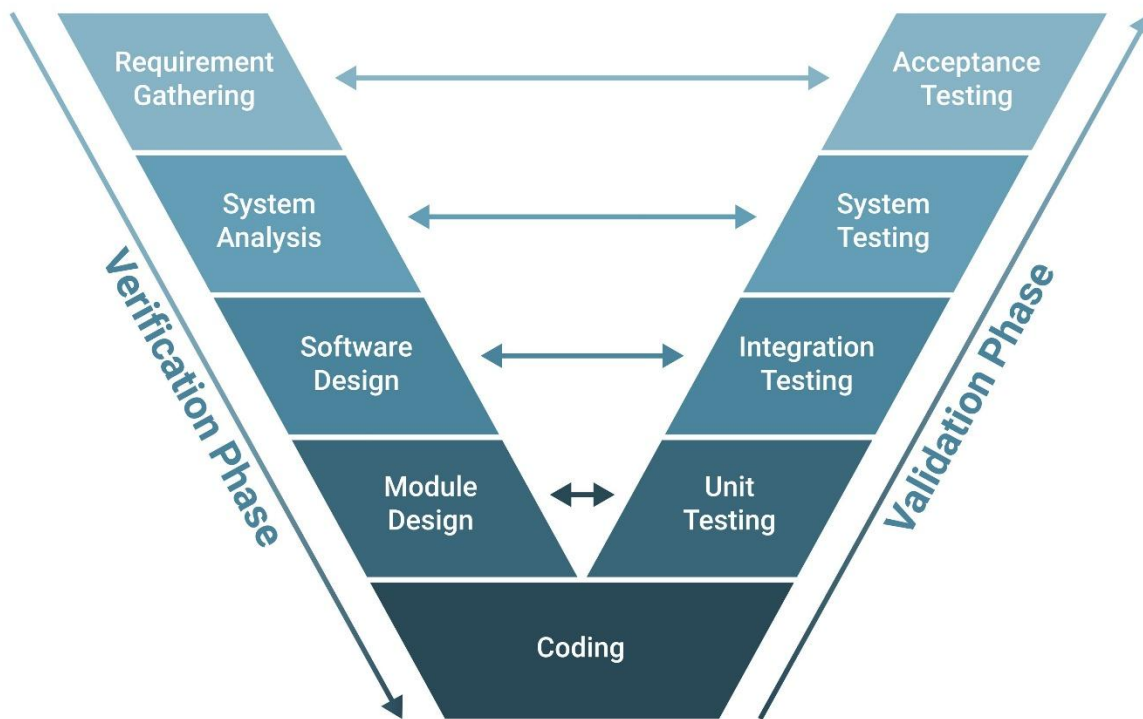


Figure 4.1: The V-Model (Verification and Validation) process, pairing each development stage with its corresponding testing stage.

Why the V-Model was Chosen:

1. **Stage-by-Stage Testing:** Unlike the Waterfall model, the V-Model requires a "Validation Plan" for every "Development Stage." For instance, as we define the **Requirement Analysis** for the Leave Module, we simultaneously define the **Acceptance Testing** to ensure it never exceeds the 25-day limit.
2. **Fixed Requirement Alignment:** Because the District Court's rules are non-negotiable, the V-model's disciplined structure ensures that no "feature creep" occurs and that the final application perfectly mirrors the original proposal.
3. **High Data Integrity:** By focusing on verification at the design stage (System Design vs. System Testing), we ensure that the **Database Schema** correctly handles the relationships between staff profiles and posting history before a single line of code is written.

4.5 Application of the V-Model to this Project

- **Requirements Analysis to Acceptance Testing:** The project proposal and its 7 use cases serve as the basis for the final user sign-off.
 - **System Design to System Testing:** The architecture of the Windows Desktop Application is tested against the performance and security requirements.
 - **Architecture Design to Integration Testing:** Ensuring that the Staff Profile module and the Leave module communicate correctly within the SQL database.
 - **Coding to Unit Testing:** Writing individual functions for seniority sorting and testing them with sample appointment dates to ensure mathematical accuracy.
-

4.6 Alternative Process Model: The Incremental Model

4.6.1 Why the Incremental Model?

If the V-Model's rigid structure becomes a bottleneck, the **Incremental Model** allows the system to be developed in parts. Instead of waiting for the entire "V" to complete before seeing a working product, the court receives functional modules one by one.

Core Benefits for the Court Project:

- **Early Value:** The court can start using the "Staff Profile" module while the complex "Seniority Logic" is still being refined.
 - **Risk Mitigation:** If the seniority sorting algorithm proves too complex, it only affects that specific increment, not the deployment of the leave or service record modules.
 - **Easier Testing:** Each increment is tested and validated separately, making it easier to identify bugs within specific features like the 25-day leave rule.
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4.6.2 Incremental Breakdown of the Project

If we were to switch to this model, the development would be divided into three distinct "Releases":

Increment	Modules Included	Purpose
Release 1	Login + Staff Profiles	Establishes the basic digital database and security.
Release 2	Service Records + Postings	Adds professional history tracking and movement logs.
Release 3	Leave Management + Seniority	Implements the complex mathematical logic and reporting.

4.6.3 Comparison: V-Model vs. Incremental Model

Feature	V-Model (Primary)	Incremental Model (Plan B)
Risk Handling	High (Testing is linked to design).	Very High (Issues are localized to one increment).
Requirement Change	Difficult to accommodate.	Easier to adjust between increments.
Visibility	User sees the system only at the end.	User sees working parts throughout the process.

Feature	V-Model (Primary)	Incremental Model (Plan B)
Suitability	Best for very clear, non-changing rules.	Best if there is any doubt about technical complexity.

4.7 Why the V-Model Remains the First Choice

Despite the strengths of the Incremental Model, the **V-Model** is maintained as the primary choice for District Court Orakzai because:

1. **Administrative Rigor:** Judicial environments prefer a single, fully-tested deployment rather than a "staged" rollout that might confuse staff.
 2. **Fixed Legal Logic:** The HR rules (Seniority and Leave) are defined by the High Court/Provincial rules and are unlikely to change during the few months of development.
 3. **Clear Documentation:** The V-Model produces the "Pair-wise" documentation (e.g., Requirement vs. Test Case) that is standard for government software audits.
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5. Project Objectives

The primary goal is to transition the court from a paper-heavy environment to a data-driven one. Specific objectives include:

- **Automation:** Replace manual HR registers with a centralized SQL-based database.
- **Data Accuracy:** Eliminate redundancy and human error in staff identity and service records.
- **Leave Logic:** Implement the "25-day annual entitlement" rule to automate leave tracking and balance calculations.
- **Seniority Management:** Automate the complex process of ranking staff based on appointment and promotion dates.

- **Reporting:** Enable the instant generation of printable reports for judicial audits and administrative reviews.
-

6. Problem Statement and Description

The administrative office of the District Court Orakzai currently faces several bottlenecks:

- **The Register Burden:** Maintaining hundreds of staff files in physical registers leads to wear and tear over time.
 - **Tracing Complexity:** Tracking the transfer history (Postings) of an employee across multiple years requires manually searching through archived service books.
 - **Leave Calculation Errors:** Manual counting of leave days often leads to incorrect balances, impacting staff entitlements.
 - **Seniority List Delays:** Generating a seniority list for promotions is currently a slow process that requires cross-referencing multiple appointment dates and civil lists.
 - **Physical Security:** Paper records are vulnerable to loss, unauthorized access, or physical damage.
-

7. Assumptions & Dependencies

7.1 Assumptions

- **Administrative Literacy:** It is assumed that the primary user (Superintendent/Clerk) has a baseline understanding of the Windows OS environment and standard GUI navigation.
- **Power Infrastructure:** Since the system is a standalone desktop application for the Orakzai District, it is assumed that the court provides a stable power supply or UPS backup to prevent database corruption during writes.
- **Rule Consistency:** The system assumes that the **Khyber Pakhtunkhwa Judicial Service Rules** regarding leave and seniority will remain constant throughout the development lifecycle.

7.2 Dependencies

- **Software Environment:** The application's functionality is dependent on the presence of **.NET Desktop Runtime 6.0** and **Microsoft SQL Server Express** on the host machine.

- **Data Migration:** The accuracy of the seniority lists depends entirely on the correct manual entry of historical data from the physical **Service Books** into the new digital database.
-

8. Project Scope

The scope of the HR Management System is strictly defined to handle the administrative needs of the Orakzai District Court.

What is Included (Functional Scope):

- **Authentication:** A secure login portal for the authorized administrative user.
- **Staff Profile Management:** Digitizing personal, contact, and geographic data (CNIC, Domicile, etc.).
- **Service & Status Tracking:** Managing designations, BPS levels (1–22), and employment types (Permanent/Contract).
- **Posting & Transfer History:** A chronological log of every station where a staff member has served.
- **Leave Management System:** A specialized module to track leave availed versus the 25-day yearly limit.
- **Seniority List Engine:** A sorting algorithm to rank staff by appointment/promotion dates.
- **Report Generation:** Specialized output formats for individual profiles and seniority lists.

What is Excluded (Non-Functional Scope):

- **Role-Based Access Control (RBAC):** As per the proposal, no tiered access is required; everyone with a login has full admin rights.
 - **Web Accessibility:** The system is a standalone desktop application and will not be accessible via the internet to ensure maximum local data security.
 - **Payroll Integration:** Financial accounting and salary disbursement are outside the current scope.
-

9. Detailed Functional Scope

Based on the specific data requirements for **District Court Orakzai**, the system will manage the following data clusters:

9.1 Identity Records

- **Personal Data:** Management of Personal Number, Name, Father's Name, CNIC, and Date of Birth.
 - **CNIC:** Must be exactly 13 digits (numeric) without dashes. The system will block any entry that does not meet this length.
 - **Date of Birth:** The system will validate that the staff member is at least **18 years of age** based on the current system date at the time of entry.
 - **Automated Superannuation Tracking:** The system shall automatically calculate the **Date of Retirement** by adding 60 years to the staff member's Date of Birth. This ensures the administration can generate "Upcoming Retirement" reports for advance planning.
 - **Personal Number:** Must be unique; the system will perform a real-time database check to prevent duplicate entries for a single employee.
- **Geographic & Contact Info:** Tracking of Domicile, Mobile Number, Email, and Permanent/Present Addresses.
- **Demographics:** Recording of Gender, Religion, and Academic Qualifications.

9.2 Service & Professional History

- **Employment Status:** Management of current Designation, BPS Level (1–22), and Status (Active, Retired, or Suspended).
- **Key Dates:** Automated tracking of Date of First Appointment and Date of Current Promotion.
- **Posting Log:** A comprehensive history of every station the staff member has served at, including the reason for each transfer.

9.3 Leave & Seniority Logic

- **Automated Leave Balance:** A dedicated module to calculate "Availed" vs. "Remaining" days based on the mandatory **25-day annual entitlement**.
- **Seniority Ranking:** An algorithm to sort staff members within their respective designations based on their appointment and promotion dates.

To eliminate human bias and grievances, the system automates seniority ranking based on the following criteria:

- **Legal Framework:** The ranking algorithm strictly follows the **Khyber Pakhtunkhwa (KP) Judicial Service Rules** for administrative staff.
 - **Priority Hierarchy:** In the event of a dispute or tie, the "**Date of First Appointment**" is given primary priority over the "**Date of Current Promotion**".
 - **Tie-Breaking:** If appointment dates are identical, the system will use the **Selection Merit Number** (assigned during recruitment) as the secondary tie-breaker. If the merit number is also identical, the system will use the staff member's age (**Age**), with the older staff member ranked higher, mirroring official KP court protocols.
-

10. Hardware & Software Requirements

To ensure the system remains functional within the court's environment, the following specifications are required.

10.1 Hardware Requirements

The system is designed to be lightweight so it can run efficiently on existing court hardware.

- **Processor:** Intel Core i3 (9th Gen) or equivalent AMD processor (Minimum: Dual-core 2.0 GHz).
- **RAM:** 4 GB (Recommended: 8 GB for faster report generation).
- **Storage:** 20 GB of available hard disk space (SSD preferred for database speed).
- **Display:** 1366 x 768 resolution or higher to properly view the Seniority List DataGrids.
- **Peripherals:** Laser printer (for printing HR reports and service history).

10.2 Software Requirements

As a standalone Windows application, the software requires a specific environment to manage the local SQL database.

- **Operating System:** Windows 10 or Windows 11 (64-bit).
- **Framework:** .NET Desktop Runtime 6.0 or higher.
- **Database Engine:** Microsoft SQL Server Express Edition (LocalDB).
- **PDF Reader:** Any standard PDF viewer (for reviewing generated HR reports).

11. Future Enhancements

While the current version meets all immediate administrative needs, the system is designed to be scalable for future court requirements:

- **Biometric Integration:** Linking the leave management system to biometric attendance devices for real-time tracking.
 - **Payroll Module:** Automating salary disbursement based on BPS levels and active service status.
 - **Web Dashboard:** A read-only web interface for the District & Sessions Judge to review real-time HR statistics from their chamber.
-

12. Requirement Specification

The requirement specification defines the detailed functional and non-functional needs of the **District Court Orakzai HR Management System**. This section serves as the technical contract between the developer and the judicial administration to ensure all operational bottlenecks are addressed.

12.1 Functional Requirements

Functional requirements describe the specific actions and behaviors the system must perform to handle the court's HR workflows.

Requirement ID	Requirement Name	Description
FR-01	User Authentication	The system must provide a secure login interface for the Authorized Admin to prevent unauthorized access to judicial staff data.

Requirement ID	Requirement Name	Description
FR-02	Profile Management	The system must allow for the creation, updating, and viewing of staff personal details, including CNIC, domicile, and contact information.
FR-03	Service Record Tracking	The system must maintain a digital record of designations, BPS levels (1–22), and employment status (Active/Retired).
FR-04	Posting & Transfer Log	The system must record a chronological history of staff station assignments and the administrative reasons for each transfer.
FR-05	Automated Leave Logic	The system must calculate availed leave and remaining balances based on the mandated 25-day annual entitlement .
FR-06	Seniority List Engine	The system must automatically sort and rank employees within their designations based on appointment and promotion dates.
FR-07	Report Generation	The system must generate formatted, printable reports for individual profiles, service histories, and seniority lists.

12.2 Non-Functional Requirements (NFRs)

Non-functional requirements specify the quality attributes and constraints of the system, ensuring it is reliable and usable in a professional court environment.

12.2.1 Security & Reliability

- **Access Control:** All administrative features must be protected by password-based authentication to ensure data privacy.
- **Data Integrity:** The system must use a local SQL database with transaction management to prevent data loss during power fluctuations. The system must prevent the deletion of a staff member if they have active "Posting History" or "Leave Records" attached to their ID.
- **Validation:** The system must validate all user inputs (e.g., CNIC format, date ranges) to prevent the entry of corrupted or illogical data.
 - **Designation-BPS Mapping:** The system must validate that the selected **BPS Level** (1–22) is legally compatible with the staff member's **Designation** (e.g., preventing a Naib Qasid from being assigned BPS-17).
 - **Chronological Date Sequencing:** The system shall block any entry where a **Transfer/Posting Date** precedes the staff member's **Date of First Appointment** to ensure historical data integrity.
- **Reliability:** The database must use ACID-compliant transactions to ensure that staff leave balances are never incorrectly deducted during a system crash.
 - **Manual Database Backup (Reliability):** To protect against local hardware failure at the Orakzai District, the system shall allow the Authorized User to trigger a manual backup of the SQL database to an external storage device.

12.2.2 Usability & Performance

- **User Interface:** The application must utilize standard Windows GUI elements to minimize training time for non-technical court clerks.
- **Response Time:** Database queries for seniority lists or report generation should complete within 2 seconds for a standard staff database.
- **Error Handling:** The system must provide clear, human-readable error messages for troubleshooting and guidance.

12.2.3 Portability & Sustainability

- **Operating System:** The system must be fully compatible with **Windows 10 and Windows 11** (64-bit) architectures.
 - **Standalone Operation:** The application must function entirely offline to comply with local data security protocols.
-

12.3 System Constraints

- **Platform Dependency:** The system is restricted to the Windows desktop environment and will not have web or mobile accessibility.
- **Administrative Scope:** The system is designed for a single-user environment where the Superintendent/Admin has full rights; tiered permissions are not required.

13.1 Use Case Diagram



Figure 13.1: Use case diagram showing the Authorized User (Superintendent/Admin) and the seven core HR management use cases.

13.2 Use Cases

13.2.1 UC01: Login to the system

Field	Details
Primary Actor	Authorized User (Admin) / Superintendent (System Administrator)
Type	Primary
Purpose	To authenticate identity and restrict access to records via password protection.
Pre-conditions	The application is installed on a Windows OS and the local database is active.
Post-conditions	The user is granted access to the main dashboard.
Alternative Flows	Invalid Credentials: If the username/password do not match the database, the system displays an error and denies access.

Field	Details
Overview	Currently, staff records at the District Court are stored in physical file cabinets or basic digital files that lack restricted access. Anyone with physical access to the office can potentially view sensitive personnel data. This use case replaces that manual vulnerability with a secure digital gateway, ensuring only those with a password can manage court records.

1. High-Level Use Case

Description: A Superintendent (System Administrator) provides credentials to gain access to the HR Management Application's features and data.

2. Essential Use Case

User Action	System Responsibility
1. Superintendent (System Administrator) /Admin requests access to the system.	2. System requests identification credentials.
3. Admin provides unique credentials.	4. System validates credentials against stored records.
	5. System grants access to HR management functions.

3. Real Use Case

- **Step 1:** The Admin launches the **Windows Desktop Application**.
- **Step 2:** The system displays a Login Window with textboxes for "**Username**" and "**Password**."
- **Step 3:** The Admin types their credentials and clicks the "**Login**" button.

- **Step 4:** The system executes a **SELECT** query against the Local SQL Database to find a matching pair.
 - **Step 5:**
 - **If match found:** The system closes the Login Window and opens the **Main Dashboard**.
 - **If no match:** The system displays a Windows Message Box stating **"Invalid Credentials."**
-

13.2.2 UC02: Manage Staff Profiles

Field	Details
Primary Actor	Authorized User (Admin)/ Superintendent (System Administrator)
Type	Primary
Purpose	To create and maintain unique records for each staff member using their Personal Number, replacing manual files with an accurate and a searchable database of personal and contact information
Pre-conditions	The Admin is successfully logged into the system.
Post-conditions	The staff member's personal and contact data is stored in the local database.

Field	Details
Alternative Flows	1. Record Not Found: If the ID does not exist, the system clears fields for a new entry. 2. Validation Error: If the CNIC format is incorrect, the system blocks the save to ensure data integrity.
Overview	At present, the court maintains personal details (CNIC, Religion, Marital Status) through manual registers. If an administrator needs to find an emergency contact or a specific personal number, they must manually flip through paper files, which is time-consuming and difficult to manage efficiently . This use case digitizes those records into a searchable format.

1. High-Level Use Case

Description: The Superintendent (System Administrator) creates, views, or modifies an employee's comprehensive profile, including their unique personal number, identity details, and contact information.

2. Essential Use Case

User Action (Intent)	System Responsibility
1. Identify the specific staff member to manage.	2. Retrieve existing profile data or provide a blank registration template.

User Action (Intent)	System Responsibility
3. Provide or update the staff member's identity markers (Name, CNIC, etc.).	4. Ensure the data matches the required identity format.
5. Provide or update contact and geographic details (Address, Domicile, etc.).	6. Commit the information to the permanent record.
7. Request a summary of the profile for verification.	8. Present the complete staff profile for review.

3. Real Use Case

- **Step 1:** The Admin/ Superintendent clicks the "**Staff Profiles**" button on the main navigation menu.
- **Step 2:** The Admin enters the **Personal Number** into the **ID search** field.
- **Step 3:** The Admin fills out the **Personal Information** section, entering data into textboxes for **Name**, **Father's Name**, **CNIC**, **Cell Number**, and **Email Address**.
- **Step 4:** The Admin selects the correct options from dropdown menus for **Gender**, **Religion**, and **Marital Status**.
- **Step 5:** In the **Contact Details** tab, the Admin types the **Present Address**, **Permanent Address**, and **Emergency Contact Number**.
- **Step 6:** The Admin enters geographic data into fields for **Domicile**, **District**, and **Tehsil**.
- **Step 7:** The Admin clicks the "**Save Profile**" button.
- **Step 8:** The system validates that the **CNIC** and **Personal Number** are present, then executes an **INSERT** or **UPDATE** query to the local SQL database.
- **Step 9:** A Windows Dialog Box appears confirming: "**Staff Profile Successfully Updated.**"

This Use Case is a "Pre-condition" for almost every other task. You cannot "Manage Leave" or "Assign Postings" if the staff profile does not exist first.

13.2.3 UC03: Manage Service Records

Field	Details
Primary Actor	Authorized User (Admin)/ Superintendent (System Administrator)
Type	Primary
Purpose	To maintain current data regarding the staff member's professional level, designation, and employment type.
Pre-conditions	The Admin is logged in and the specific Staff Profile already exists in the system.
Post-conditions	The employee's service record is updated and becomes available for seniority calculations.
Alternative Flows	Invalid Designation: If the Admin enters a designation not found in the official court list, the system prompts the user to select a valid role.
Overview	Currently, an employee's professional standing—such as their BPS grade, department, and whether they are permanent or contract—is tracked via

Field	Details
	manual service books . Cross-referencing these details for different staff members is slow. This use case centralizes professional data to allow for instant verification of a staff member's current status within the court hierarchy.

1. High-Level Use Case

Description: The Admin views or updates the professional designation, BPS level, and employment nature of an existing staff member.

2. Essential Use Case (Logical Flow)

User Action (Intent)	System Responsibility
1. Provide the unique staff identifier.	2. Retrieve and display the current service status.
3. Modify professional details (e.g., Change BPS level or Designation) .	4. Ensure the updated designation is part of the authorized court staff list .
5. Confirm the changes.	6. Update the professional record in the staff archive.

3. Real Use Case (Technical Flow)

- **Step 1:** The Admin/ Superintendent (System Administrator) navigates to the "Service Info" section of the desktop application.
- **Step 2:** The Admin enters the **Personal Number** and clicks "**Load Service Data**".

- **Step 3:** The system fetches the current **Designation**, **BPS Level**, and **Department** from the SQL database and populates the textboxes.
 - **Step 4:** The Admin selects either "**Permanent**" or "**Contract**" using a radio button group.
 - **Step 5:** The Admin updates the **BPS Level** (e.g., 1 to 22) using a dropdown menu selector.
 - **Step 6:** The Admin clicks the "**Update Service Record**" button.
 - **Step 7:** The system executes a SQL UPDATE command on the service history table.
 - **Step 8:** A **Windows notification** confirms that the professional status has been updated.
-

13.2.4 UC04: Manage Postings & Transfers

Field	Details
Primary Actor	Authorized User (Admin) / Superintendent (System Administrator)
Type	Primary
Purpose	To document staff movement, maintain a searchable posting history, and record reasons for transfers.
Pre-conditions	The Admin is successfully authenticated and the staff member's basic profile exists in the system.

Field	Details
Post-conditions	The current posting is archived into the history, and the new assignment is set as the active station in the database.
Alternative Flows	Date Conflict: If a transfer date is entered that is earlier than the current posting's start date, the system prompts for a correction.
Overview	In the current manual system, tracking where a staff member has been posted over several years involves searching through multiple paper service books. Determining the reason for a specific transfer or verifying a previous station is often delayed by physical record retrieval. This use case digitizes the movement of staff, allowing the court to instantly view a chronological history of assignments and the administrative reasons behind each transfer.

1. High-Level Use Case

Description: The Admin/ Superintendent (System Administrator) records the transfer of an employee to a new station, specifying the date and the reason for the change.

2. Essential Use Case (Logical Flow)

User Action (Intent)	System Responsibility
1. Provide staff identifier.	2. Retrieve current assignment details.
3. Input new station and transfer date.	4. Validate that the date aligns with the staff member's service timeline.

User Action (Intent)	System Responsibility
5. Provide reason for the transfer.	6. End the current posting record.
7. Commit transfer data.	8. Create a new "Current Posting" record and update the "Posting History".

Real Use Case (Technical Flow)

- **Step 1:** The Admin / Superintendent navigates to the "**Postings & History**" tab in the Windows desktop application.
 - **Step 2:** The Admin enters the Personal Number and clicks the "**Search**" icon.
 - **Step 3:** The system fetches and displays the **Current Place of Posting** and **Start Date** from the database.
 - **Step 4:** The Admin types the **New Station** name into a textbox.
 - **Step 5:** The Admin selects the **Transfer Date** using a Windows calendar control.
 - **Step 6:** The Admin selects a **Reason for Transfer** (e.g., **Promotion, Administrative, Request**) from a dropdown list.
 - **Step 7:** The Admin clicks the "**Execute Transfer**" button.
 - **Step 8:** The system runs a SQL transaction that **updates** the existing posting record with an **End Date** and **inserts** the new station into the **Current_Posting** table.
 - **Step 9:** A confirmation message appears: "**Transfer successful; Posting history updated**".
-

13.2.5 UC05: Manage Leave Records

Field	Details
Primary Actor	Authorized User (Admin)/ Superintendent (System Administrator)
Type	Primary
Purpose	To automate the tracking of annual leave (25 days per staff) and maintain an accurate history of leave availed .
Pre-conditions	The Admin is authenticated, and the employee's personal record is already established in the database .
Post-conditions	The Remaining Balance is decremented, and a new entry is added to the year-wise leave history .
Alternative Flows	Insufficient Balance: If the requested days exceed the current remaining balance, the system blocks the transaction and alerts the Admin.
Overview	Under the current manual system, each staff member's leave is tracked in individual paper accounts. Calculating the remaining balance from the 25-day annual entitlement and cross-referencing year-wise history is a slow, error-prone process. This use case automates the calculation logic, ensuring that no staff member exceeds their allowed leave and providing an instant, accurate record of leave availed and remaining balance.

1. High-Level Use Case

Description: The Admin/ Superintendent enters the number of leave days for a staff member, and the system validates this against the 25-day annual limit before updating the remaining balance and history.

2. Essential Use Case (Logical Flow)

User Action (Intent)	System Responsibility
1. Identify the staff member.	2. Retrieve current leave balance (based on 25-day annual entitlement) .
3. Input the number of days for the current leave request.	4. Validate that Requested Days does not exceed Remaining Balance.
5. Confirm the leave entry.	6. Subtract requested days from the balance and record the date and history .

3. Real Use Case (Technical Flow)

- **Step 1:** The Admin/ Superintendent opens the "**Leave Management**" module within the Windows desktop application.
- **Step 2:** The Admin enters the Personal Number and clicks "**Check Entitlement**".
- **Step 3:** The system queries the local database and displays the "**Days Availled**" and "**Days Remaining**" in UI labels.
- **Step 4:** The Admin types the number of days into the "**Apply Leave**" textbox and selects the start/end dates.
- **Step 5:** The Admin clicks the "**Process Leave**" button.
- **Step 6:** The system executes a logic check: **IF (input_days <= current_balance)** THEN proceed **ELSE** display "**Insufficient Balance**" error.

- **Step 7:** Upon validation, the system executes a SQL transaction to **UPDATE** the balance and **INSERT** a record into the **LeaveHistory** table.
 - **Step 8:** A confirmation pop-up displays: "**Leave recorded successfully. New balance: [X] days.**"
-

13.2.6 UC06: Manage Seniority Lists

Field	Details
Actor	Authorized User (Admin) / Superintendent (System Administrator)
Type	Primary
Purpose	To calculate and maintain accurate seniority rankings based on appointment and promotion dates.
Pre-conditions	The Admin is authenticated, and all relevant staff members have their appointment dates entered in their service records.
Post-conditions	A designation-wise list of staff is generated, ranked from most senior to most junior.
Alternative Flows	Missing Date of Appointment: If a staff member's appointment date is missing, the system excludes them from the ranking and prompts the Admin to update their service record.

Field	Details
Overview	In the current manual environment, determining seniority is a complex task that involves manually comparing the appointment and promotion dates of dozens of staff members across different registers. This process is susceptible to human error, which can lead to grievances regarding promotions or administrative privileges. This use case automates the ranking logic, ensuring that seniority lists are instantly generated and perfectly sorted according to official court rules .

1. High-Level Use Case

Description: The Admin/ Superintendent selects a specific staff designation, and the system automatically generates a list of all relevant employees ranked by their seniority.

2. Essential Use Case (Logical Flow)

User Action (Intent)	System Responsibility
1. Select the staff group (designation) to be ranked.	2. Filter the database for all active employees within that designation.
3. Request seniority calculation.	4. Sort the filtered staff based on Date of Appointment (primary) and Date of Promotion (secondary).
	5. Assign a numerical seniority rank to each member.
	6. Present the finalized seniority list for review.

3. Real Use Case (Technical Flow)

- **Step 1:** The Admin/ Superintendent clicks the "**Seniority Management**" icon on the dashboard.
 - **Step 2:** The Admin selects a Designation (e.g., "**Clerks**" or "**Stenographers**") from a dropdown menu.
 - **Step 3:** The Admin clicks the "**Generate List**" button.
 - **Step 4:** The system executes a SQL query: `SELECT Name, AppointmentDate, PromotionDate FROM Staff WHERE Designation = 'SelectedValue' ORDER BY AppointmentDate ASC, PromotionDate ASC.`
 - **Step 5:** The system processes the results and displays them in a **DataGrid** view on the screen, showing the rank, name, and service dates.
 - **Step 6:** A "**Refresh**" option allows the Admin to update the list if any new service data was added during the session.
-

13.2.7 UC07: Generate HR Reports

Field	Details
Actor	Authorized User (Admin)/ Superintendent (System Administrator)
Type	Primary
Purpose	To provide formatted, printable summaries of HR data for administrative review and official documentation.
Pre-conditions	The Admin is authenticated, and the database contains the records required for the specific report type .

Field	Details
Post-conditions	A formatted report is displayed on the screen and/or sent to the printer.
Alternative Flows	No Data Found: If the Admin attempts to generate a report for a Personal Number that does not exist, the system displays a "No records found" notification.
Overview	In the current manual setup at District Court Orakzai, generating a comprehensive report—such as a staff member's full service history or a current seniority list—requires a clerk to manually find, compile, and type data from various paper registers. This process is not only time-consuming but also prone to typographical errors. This use case enables the instant generation of formatted, professional reports including individual profiles, leave histories, and seniority lists, which can be printed for official court use.

1. High-Level Use Case

Description: The Admin/ Superintendent (System Administrator) selects a specific report category and parameters (such as a staff ID or designation) to produce a formatted document for viewing or printing .

2. Essential Use Case (Logical Flow)

User Action (Intent)	System Responsibility
1. Select the desired report type (Individual Profile, Leave Record, Seniority, etc.) .	2. Prompt for necessary filters (e.g., Staff ID or Designation).
3. Provide filter criteria.	4. Retrieve all associated data from various modules (Identity, Service, Leave, Postings).
5. Request report generation.	6. Organize data into a standardized court format and present it for preview.
7. Finalize the output.	8. Send the document to the system's output device (Printer/PDF).

3. Real Use Case (Technical Flow)

- **Step 1:** The Admin/ Superintendent clicks the "**Reports & Printing**" icon on the main menu.
- **Step 2:** The Admin selects a radio button for the report type: "**Individual Profile**", "**Leave History**", or "**Seniority List**".
- **Step 3:** For individual reports, the Admin types the **Personal Number** into a search box.
- **Step 4:** The Admin clicks the "**Preview Report**" button.
- **Step 5:** The system executes a complex JOIN query across the Staff, Service, and Leave tables in the **local SQL database**.
- **Step 6:** The system populates a reporting template (like a Crystal Report or a formatted Windows Print document) with the retrieved data.
- **Step 7:** The Admin reviews the document in a **Print Preview** window.

- **Step 8:** The Admin clicks the "**Print**" button, triggering the Windows print dialogue to send the document to the connected printer.
-

13.2.8 UC08: Logout from System

Field	Details
Primary Actor	Authorized User (Admin) / Superintendent
Type	Secondary / Supporting
Purpose	To securely terminate the current administrative session and return the application to the login state.
Pre-conditions	The Admin is successfully logged into the system.
Post-conditions	The session is invalidated, the main dashboard is closed, and the login window is reopened.
Overview	To prevent unauthorized access when the Superintendent is away from the computer, this use case allows for the immediate termination of the active session without closing the entire application process.

High-Level Use Case

Description: The Superintendent selects the logout option to end their active session and restrict further access until valid credentials are provided again.

Essential Use Case

User Action	System Responsibility
1. Admin requests to log out.	2. System confirms the intent to end the session.
3. Admin confirms logout.	4. System terminates session and restricts access to HR functions.

Real Use Case (Technical Flow)

- **Step 1:** The Admin clicks the "**Logout**" button on the Main Dashboard.
- **Step 2:** The system displays a confirmation dialogue: "**Are you sure you want to log out?**".
- **Step 3:** The Admin clicks "**Yes.**"
- **Step 4:** The system clears any temporary session variables in memory.
- **Step 5:** The system **closes** the Main Dashboard and **re-launches** the Login Window

14.Domain Model

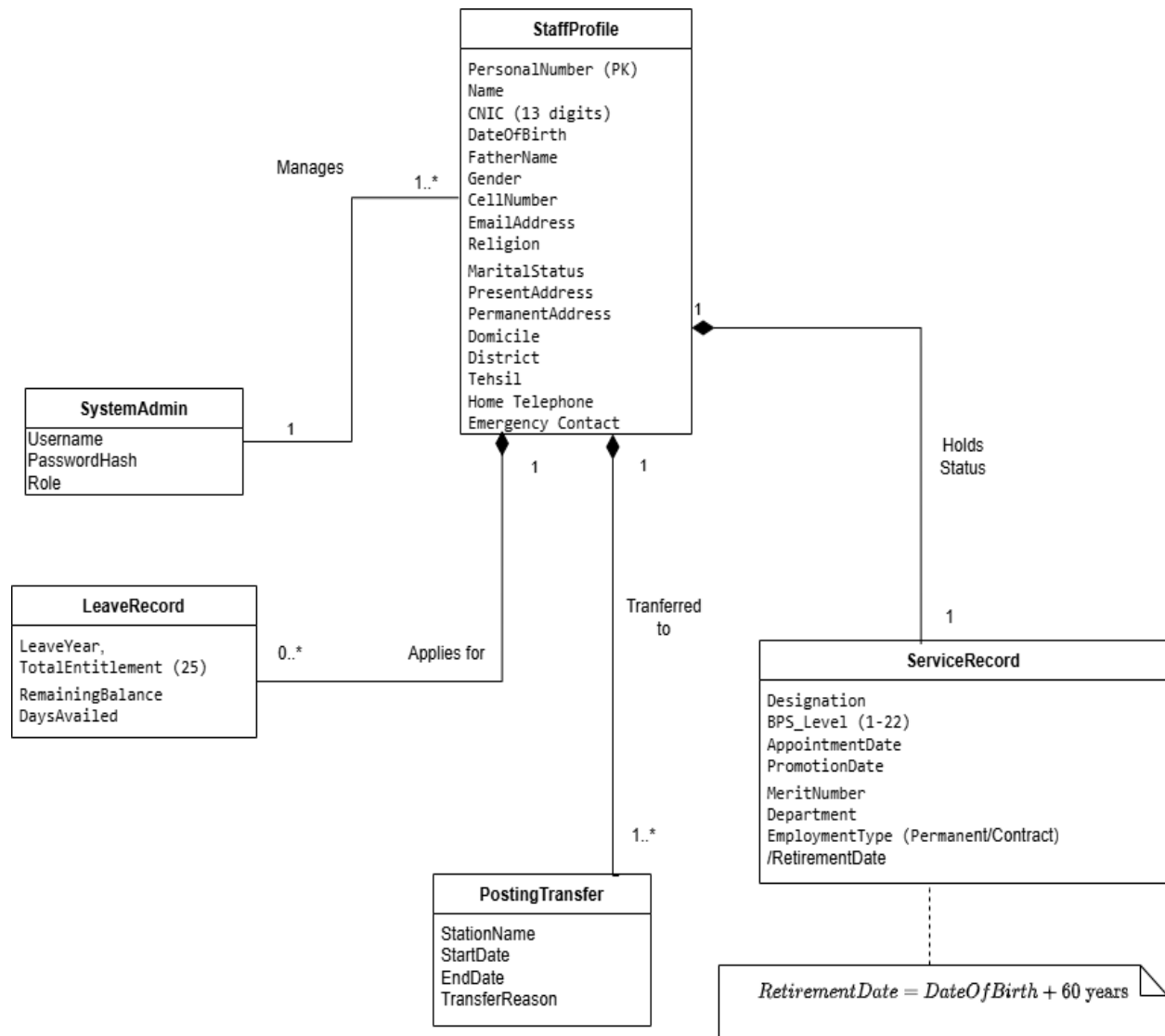


Figure 14.1: Domain model of the HR Management System, showing the core concepts and their associations.

Concept Category and Concepts Table

Concept Category	Extracted Concepts
Physical or tangible objects	Court Premises
Places	Station, District, Tehsil
Transactions / Events	Posting, Transfer, Leave Request
Roles of people	System Admin, Superintendent, Staff Member, Clerk, Computer Operator, Stenographer, Naib Qasid
Records of work, contracts, legal matters	Staff Profile, Service Record, Leave Record, Posting Transfer Record
Organizations	District Court Orakzai, Departments
Rules and policies	25-Day Leave Entitlement, Seniority Sorting Logic, KP Judicial Rules
Abstract noun concepts	Seniority Rank, Leave Balance, BPS Level, Designation, Merit Number
Descriptions of things	Personal Number, CNIC

Applying the Associations Checklist

Association Category	Identified Association	Concept A	Concept B
A uses or manages B	Admin manages staff identities	SystemAdmin	StaffProfile
A is a logical part of B	Service details are a logical part of an employee	ServiceRecord	StaffProfile
A is owned by B	Leave accounts are owned by an employee	LeaveRecord	StaffProfile
A is an event related to B	A transfer is an event related to an employee	PostingTransfer	StaffProfile
A is a description for B	Professional status describes an employee	ServiceRecord	StaffProfile
A is known/logged/recorded in B	Employee movement is logged in a transfer record	StaffProfile	PostingTransfer

15. Data Flow Diagram

15.1 Level 0 DFD

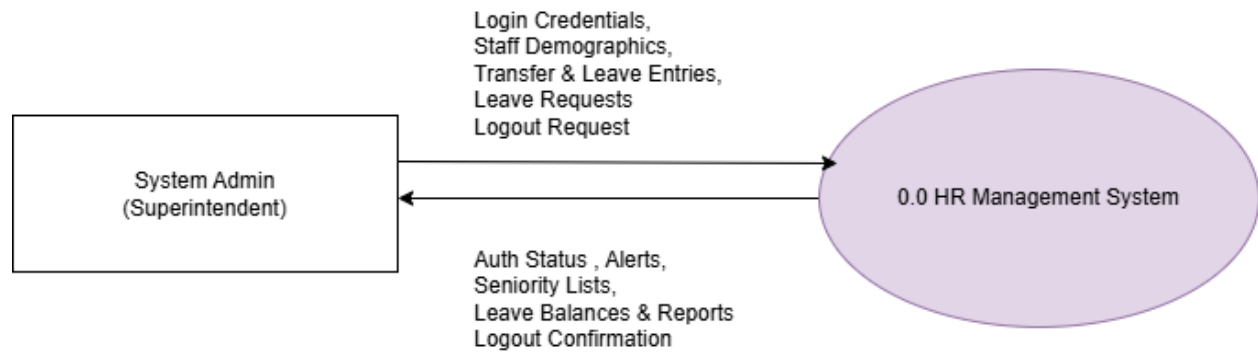


Figure 15.1: Level 0 Data Flow Diagram (context diagram) of the HR Management System.

15.2 Level 1 DFD

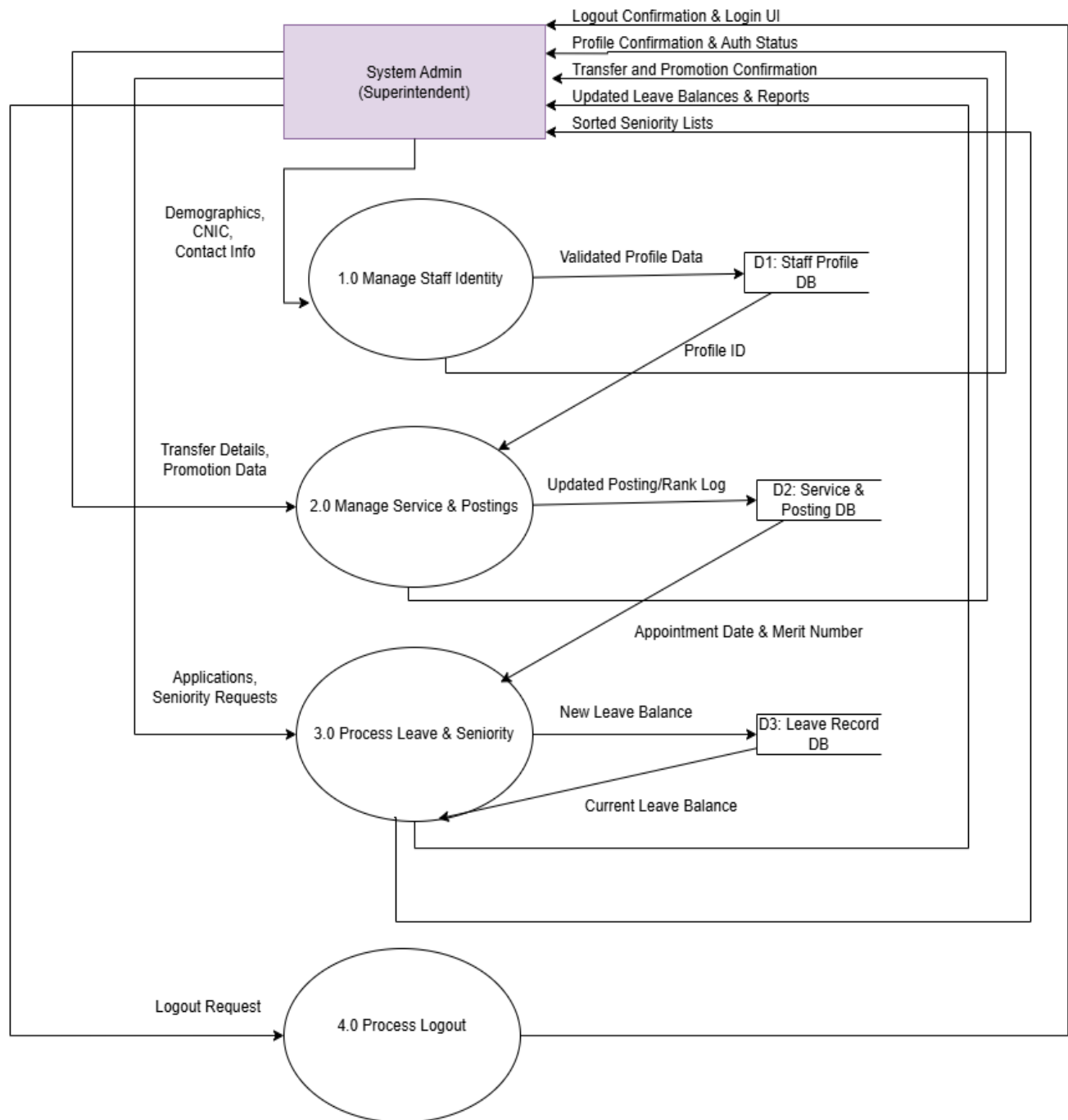


Figure 15.2: Level 1 Data Flow Diagram, decomposing the system into its major processes.

15.3 Level 2 DFD

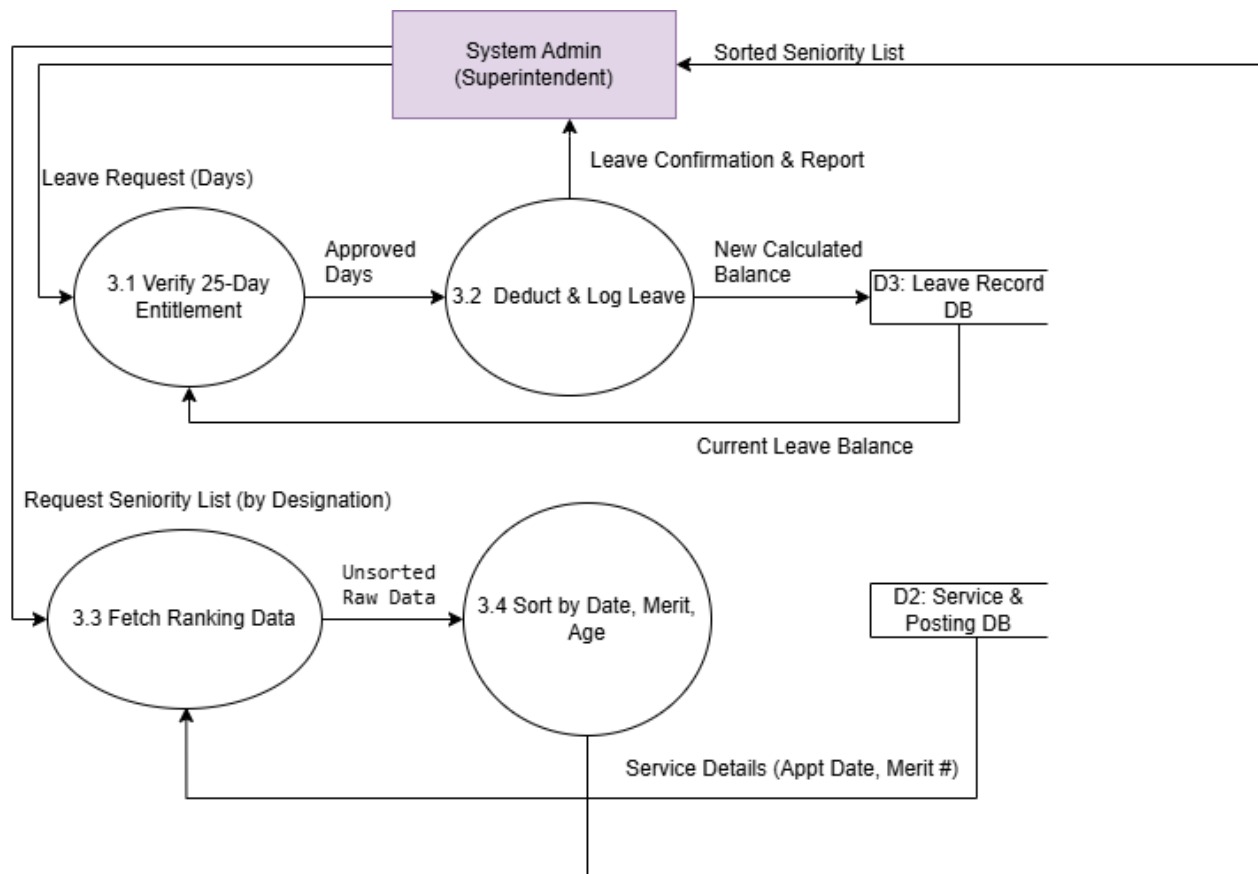


Figure 15.3: Level 2 Data Flow Diagram, detailing the internal data flows of the core processes.

16. Class Diagram

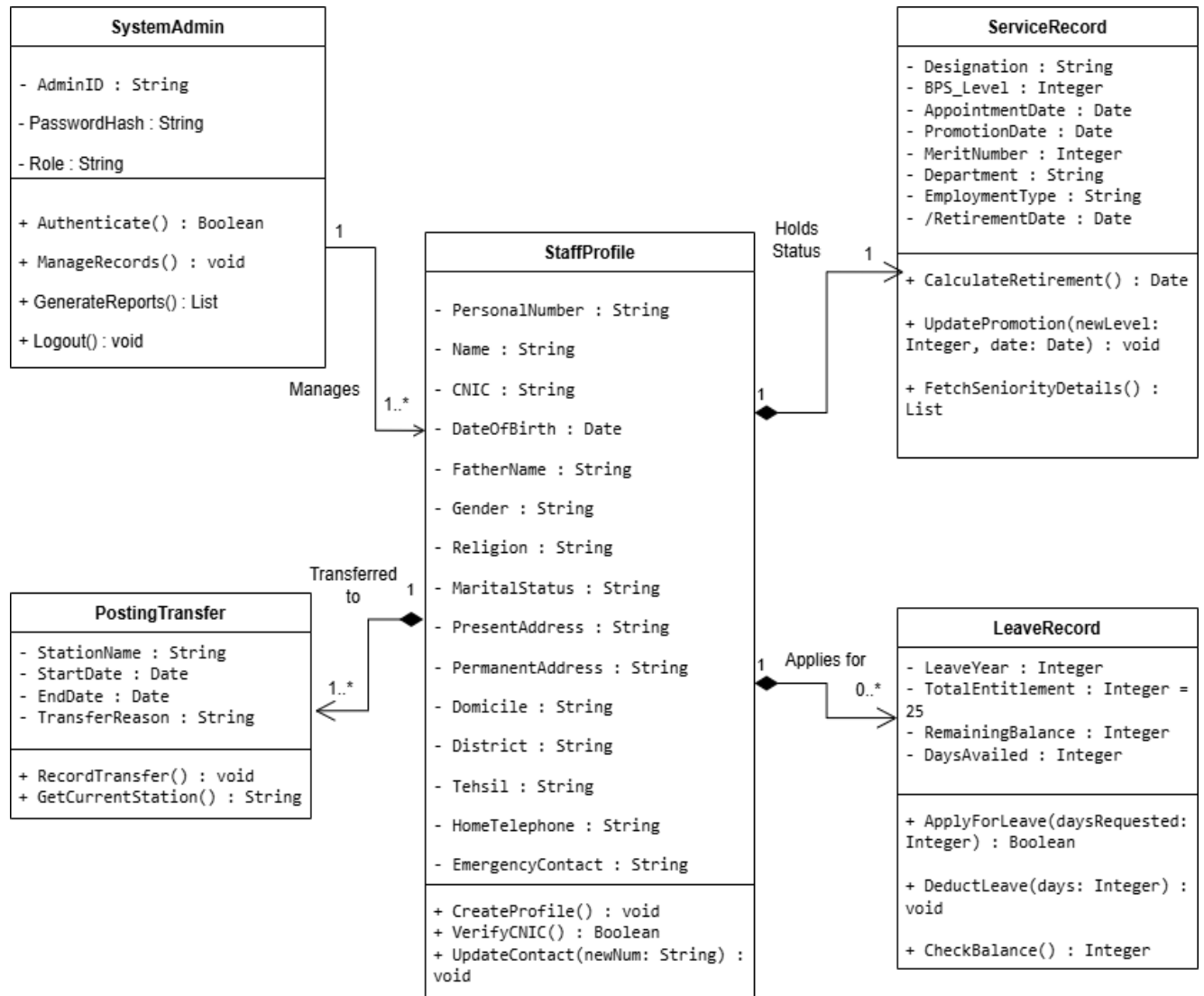


Figure 16.1: Class diagram of the HR Management System showing the principal classes and their relationships.

17. System Sequence Diagrams

17.1 SSD 1: SSD 1: Login to the System (UC01)

This diagram focuses on authentication.

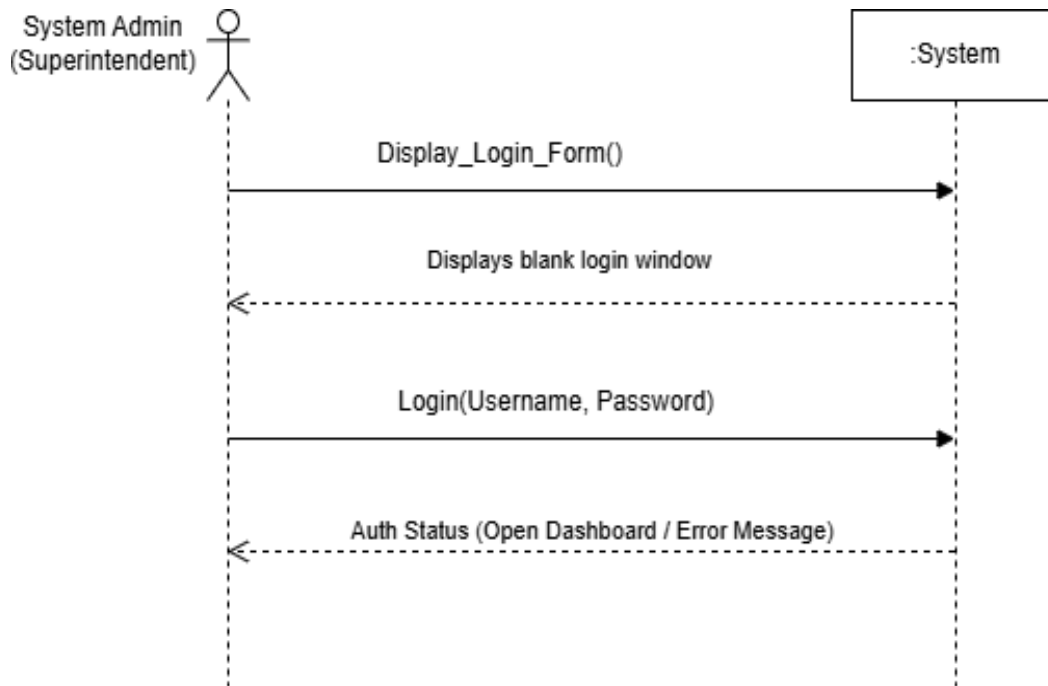


Figure 17.1: System Sequence Diagram for UC01 – Login to the System.

17.2 SSD 2: Manage Staff Profiles (UC02)

Covers creating and updating staff personal data.

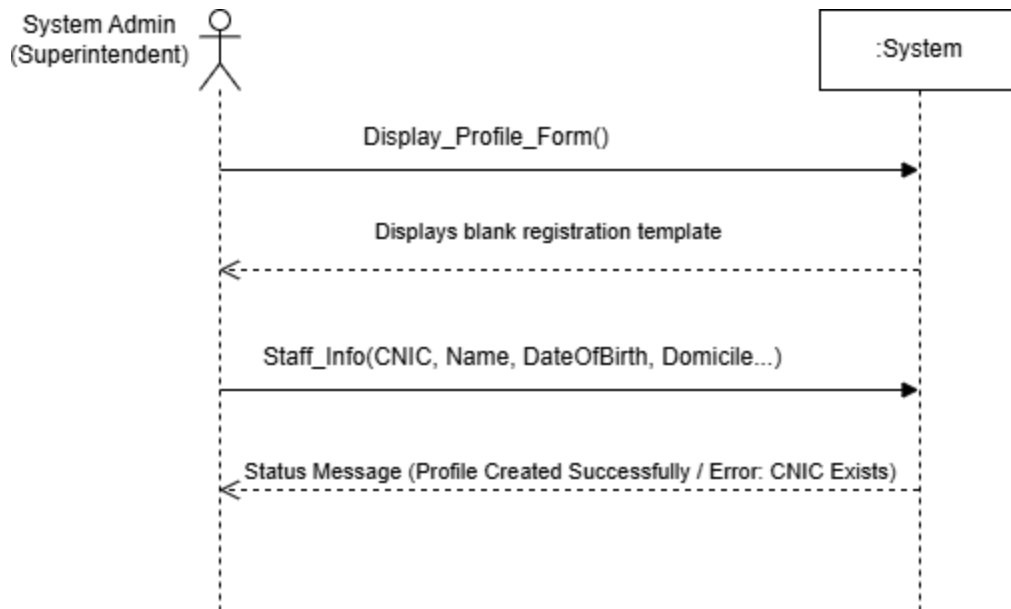


Figure 17.2: System Sequence Diagram for UC02 – Manage Staff Profiles.

17.3 SSD 3: Manage Service Records (UC03)

Focuses on professional standing and BPS grades.

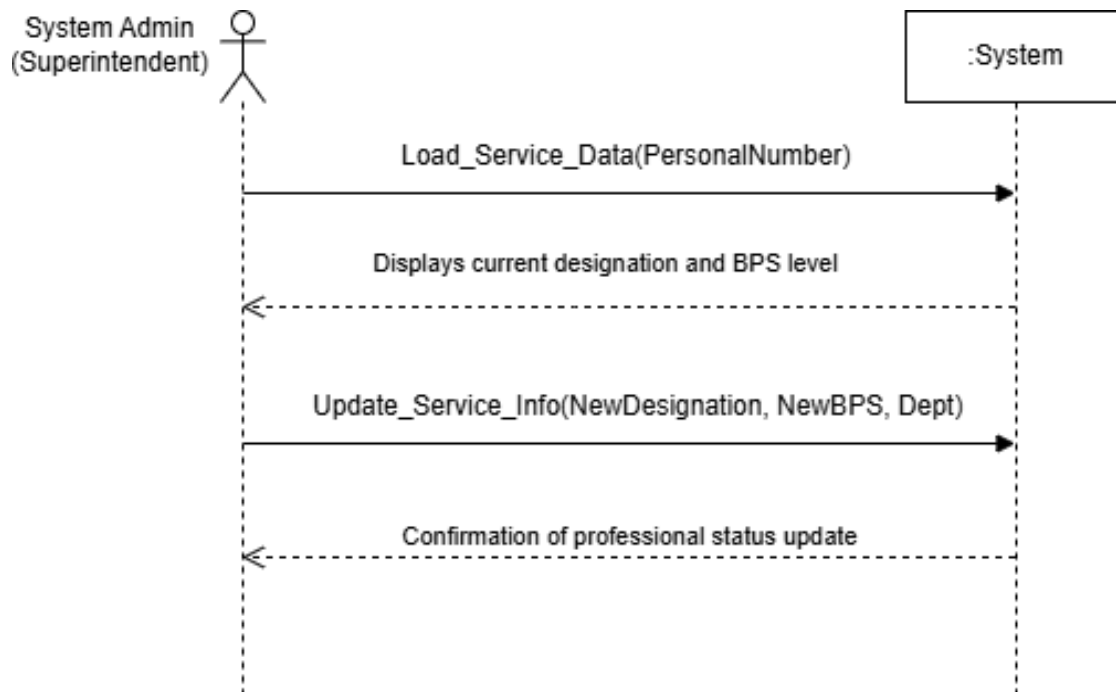


Figure 17.3: System Sequence Diagram for UC03 – Manage Service Records.

17.4 SSD 4: Manage Postings & Transfers (UC04)

Logs the movement of staff between stations.

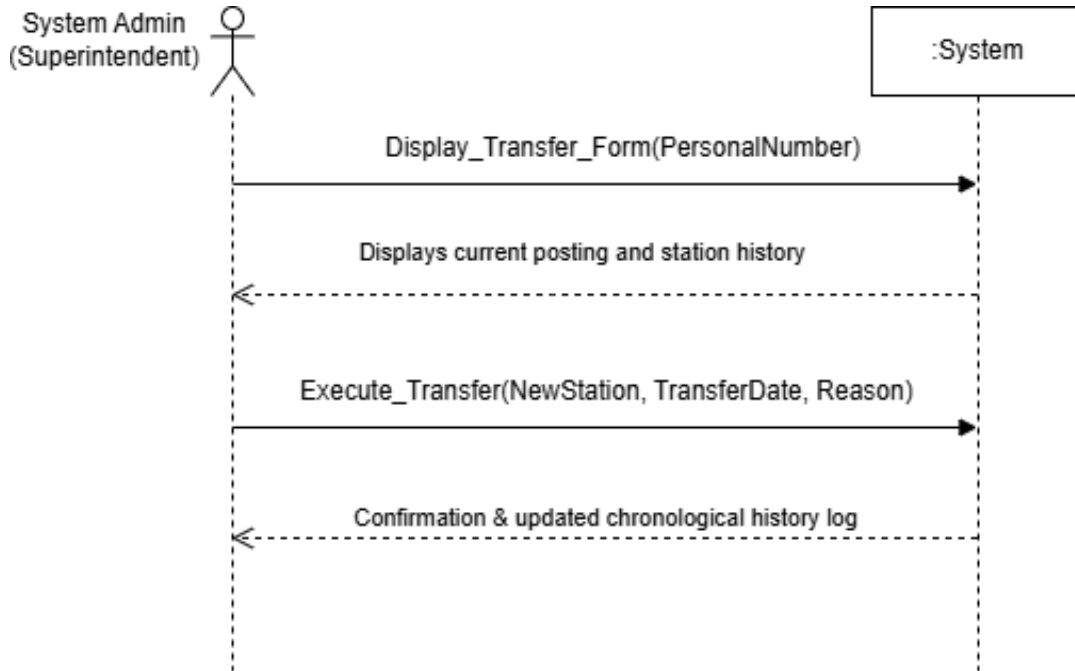


Figure 17.4: System Sequence Diagram for UC04 – Manage Postings & Transfers.

17.5 SSD 5: Manage Leave Records (UC05)

Implements the automated 25-day annual entitlement rule.

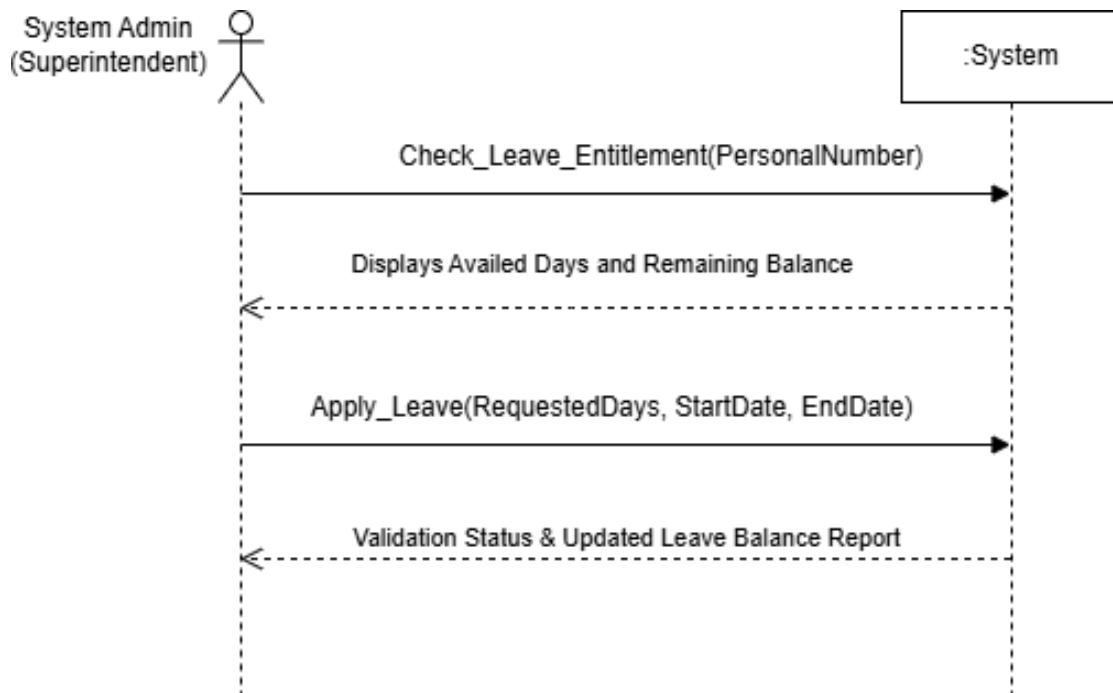


Figure 17.5: System Sequence Diagram for UC05 – Manage Leave Records.

17.6 SSD 6: Manage Seniority Lists (UC06)

Automates ranking based on KP Judicial Service Rules

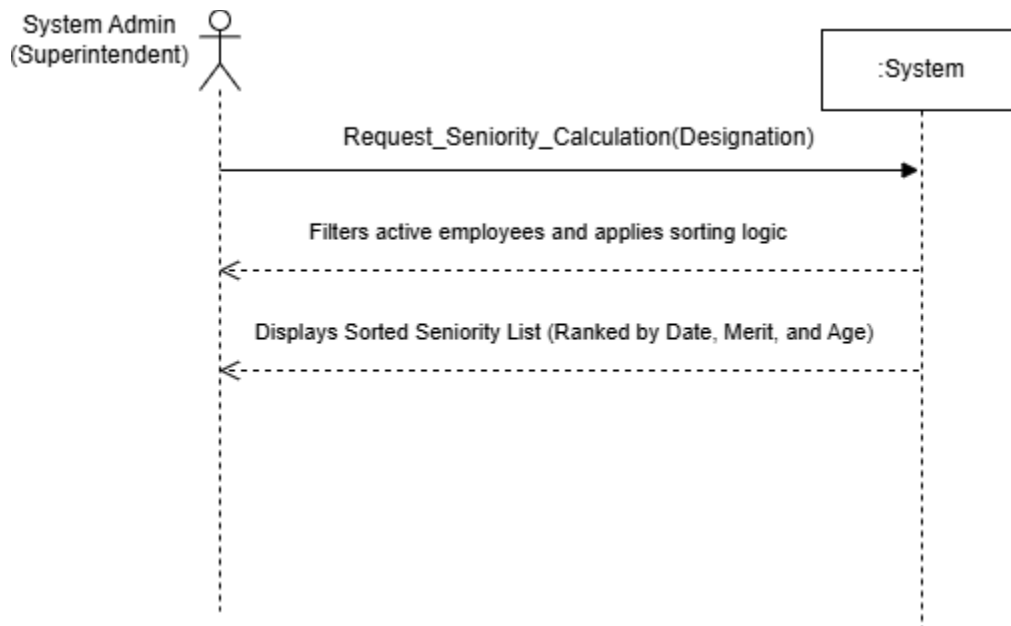


Figure 17.6: System Sequence Diagram for UC06 – Manage Seniority Lists.

17.7 SSD 7: Generate HR Reports (UC07)

Produces formatted summaries for judicial review.

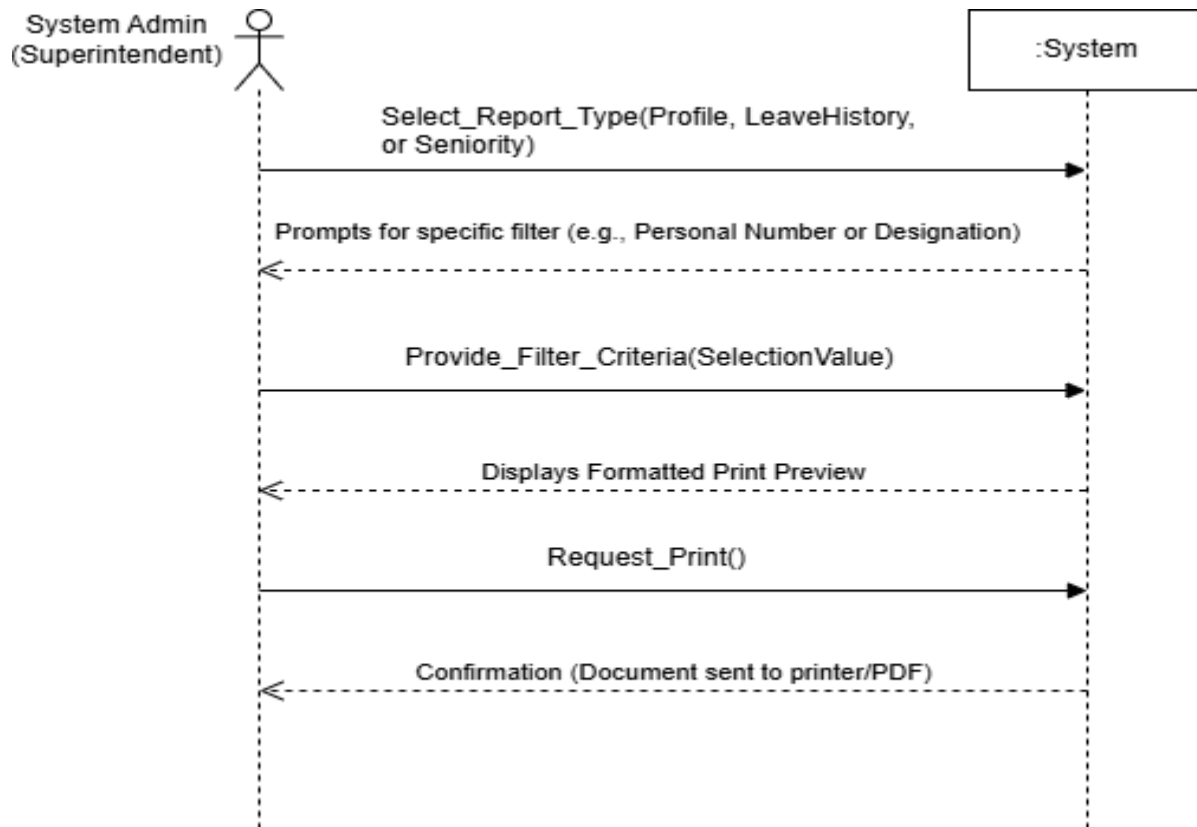


Figure 17.7: System Sequence Diagram for UC07 – Generate HR Reports.

17.8 SSD 8: Logout from System (UC08)

To securely terminate the current administrative session and return the application to the login state.

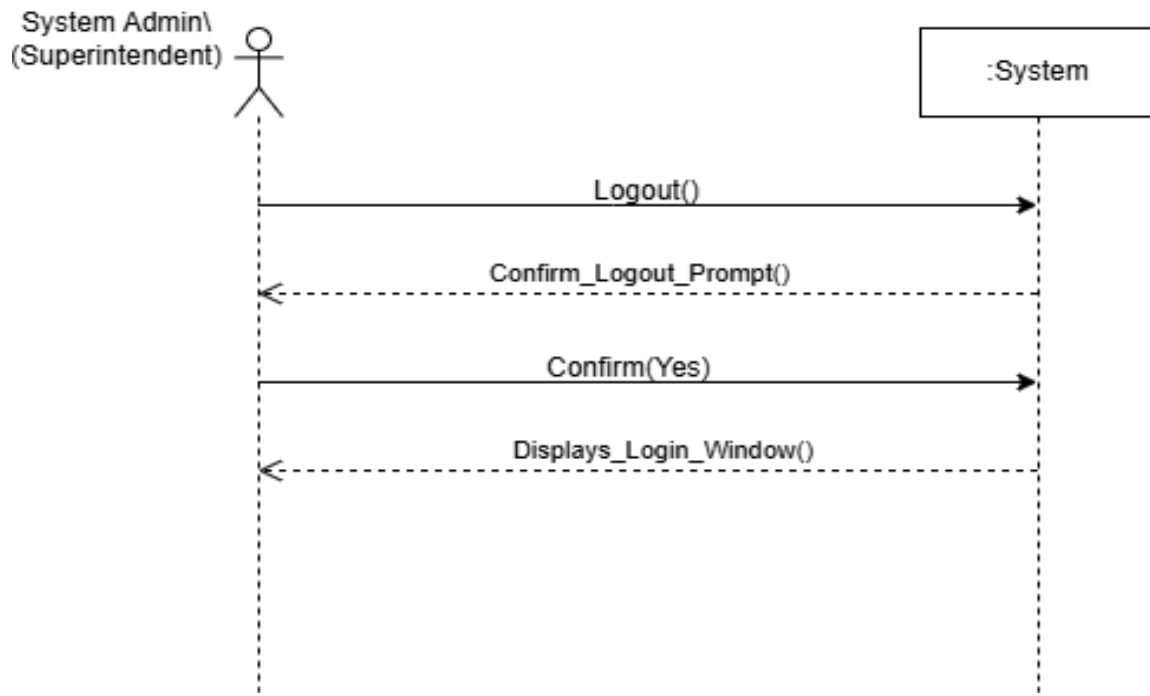


Figure 17.8: System Sequence Diagram for UC08 – Logout from System.

18. Contracts

18.1. FOR SSD1(UC01) Login to the system

18.1.1 Contract 1: Login Form Initialization

Field	Description
Operation	Display_Login_Form()
Type	UI Window Controller / SessionManager
Cross References	UC01: Login to the system, FR-01: User Authentication

Field	Description
Responsibility	Initializes and renders the desktop security gateway window to prevent unauthorized application access.
Pre-conditions	The application executable is launched on a Windows OS environment and the local SQL Server Express database connection is active.
Post-conditions	An instance of the user interface (Login Window) was initialized and rendered on the host screen. Input textboxes for Username and Password were exposed to user focus. No changes occurred to persistent database entities or state memory variables.
Exceptions	DatabaseConnectionException: Raised if the local SQL Server Express instance is unreachable, popping up a critical system recovery alert block.
Output	UI window handle rendered onto the default local presentation device.
Notes	Serves as the primary barrier protecting sensitive judicial personnel information before any data structures are cached into volatile application RAM.

18.1.2 Contract 2: User Identity Authentication

Field	Description
Operation	Login(Username, Password)
Type	System Security Controller / SessionManager
Cross References	UC01: Login to the system, FR-01: User Authentication
Responsibility	Verifies administrator credentials using explicit queries to authorize access to underlying functional modules.
Pre-conditions	The application is actively running and displaying the unauthenticated Login Window state. The user has filled in non-empty alphanumeric strings into the credential fields.
Post-conditions	The system executed a secure SELECT query against the local database table mapping SystemAdmin parameters. Scenario A (Match Found):A validated runtime user session instance was successfully allocated in application memory. The unauthenticated LoginWindow view object was closed/disposed .The primary MainDashboard user interface view shell was initialized, bound, and displayed to the Admin. Scenario B (Match Not Found):The application rejected the transaction, maintaining a strict unauthenticated memory state .A standard Windows message box interface dialogue displaying "Invalid Credentials" was initialized and rendered on-screen.

Field	Description
Exceptions	InvalidCredentialsException: Fired if the supplied username and password string pair do not match data records within the local configuration schema.
Output	Cryptographically clear system authorization session token in application memory.
Notes	This system runs entirely offline in a single-user standalone environment for maximum local security.

18.2 SSD2, UCO2: Manage Staff Profiles

18.2.1 Contract 3: Profile Registration Form Initialization

Field	Description
Operation	Display_Profile_Form()
Type	ProfileManagementController / StaffProfile
Cross References	UC02: Manage Staff Profiles, FR-02: Profile Management

Field	Description
Responsibility	Resets and maps form inputs to clear old system variables, creating a blank template for profile registration entries.
Pre-conditions	The Authorized Admin is successfully logged into the system.
Post-conditions	An instance of the profile management graphical user interface was initialized and rendered on the monitor. Form fields, input textboxes, and dropdown navigation selectors for identity, contact, and geographic data clusters were cleared of legacy data and exposed to user focus. The system state remained unchanged with zero writes executed to the local SQL storage layer.
Exceptions	SessionTimeoutException: Dispatched if the active session tracking token has expired or been corrupted before the layout window is drawn.
Output	Rendered graphical text field layout parameters on screen.
Notes	This profile container view operates as a baseline dependency; no other transactional records can be attached to an employee if this structure is not completed first.

18.2.2 Contract 4: Profile Submission and Database Write

Field	Description
Operation	Staff_Info(CNIC, Name, DateOfBirth, Domicile...)
Type	ProfileManagementController / StaffProfile
Cross References	UC02: Manage Staff Profiles , FR-02: Profile Management
Responsibility	Validates incoming identity fields to systematically record a clean employee profile into the local database.
Pre-conditions	The Authorized Admin is successfully authenticated and interacting with an active profile entry form window. The mandatory identifier attributes (PersonalNumber, CNIC, Name) are populated with non-empty input strings.
Post-conditions	The system validated that the input CNIC string consists of exactly 13 numeric digits and that the calculated age based on the supplied DateOfBirth is $\geq$ 18 years relative to the current system calendar date. Scenario A (Valid Data & New Unique Personal Number): - A brand new StaffProfile entity instance was initialized and added to the local database using a SQL INSERT command.

Field	Description
	- The system automatically computed the superannuation milestone by adding 60 years to the validated DateOfBirth value, storing it in the record. - All associated demographic, contact, and administrative parameters were safely written to the corresponding fields in the database schema. - A success dialog notification ("Staff Profile Successfully Updated") was generated and displayed to the Admin. Scenario B (Valid Data & Existing Personal Number Match): - The existing StaffProfile instance matching the target identifier was opened for modification. - A SQL UPDATE transaction was executed to overwrite altered attributes with the newly provided values. - A success confirmation dialog was presented. Scenario C (Validation Fault - Invalid Length or Duplicate Key Collision): - The persistent database write transaction was blocked/aborted to maintain data integrity. - The current system memory state was rolled back to its immediate pre-submission layout. - An explicit Windows error message box highlighting the validation failure (e.g., "Error: CNIC Exists" or bad data formatting) was triggered on-screen.
Exceptions	InvalidCNICFormatException: Thrown if the string fails regex parsing or does not equal 13 digits. UnderageException: Fired if calculated timeline age values are less than 18.

Field	Description
	.DuplicatePersonalNumberException: Raised if a primary key violation occurs during an INSERT action.
Output	Relational data rows written to the local SQL data store block.
Notes	Superannuation date records are strictly derived using calculation formulas rather than manual text field entry fields to eliminate administrative errors.

18.3 SSD3, UC03: Manage Service Records

18.3.1 Contract 5: Service Record Loading Query

Field	Description
Operation	Load_Service_Data(PersonalNumber)
Type	ServiceHistoryController / ServiceRecord
Cross References	UC03: Manage Service Records , FR-03: Service Record Tracking
Responsibility	Queries the persistent database layer to fetch active employment parameters for user viewing.

Field	Description
Pre-conditions	The Authorized Admin is successfully logged into the desktop application. A StaffProfile instance matching the input PersonalNumber exists within the local database.
Post-conditions	The system executed a read-only lookup query against the database using the provided PersonalNumber. The active attributes for the employee's Designation, BPS_Level, and Department were fetched from the database storage layer. The graphical user interface forms and textboxes were initialized and populated with these retrieved records for user visibility. The state of the system remained unchanged, with no structural database writes performed.
Exceptions	RecordNotFoundException: Fired if the supplied identifier does not map to an existing record, spawning a text notification layout.
Output	Populated textbox interface bindings across the active form framework.
Notes	This execution block gathers foundational parameters used to construct multi-table JOINS for subsequent seniority list calculations.

18.3.2 Contract 6: Service Record Update Execution

Field	Description
Operation	Update_Service_Info(NewDesignation, NewBPS, Dept)

Field	Description
Type	ServiceHistoryController / ServiceRecord
Cross References	UC03: Manage Service Records, FR-03: Service Record Tracking
Responsibility	Applies business validation mappings to safely update the BPS level and title details of an employee.
Pre-conditions	The Authorized Admin is authenticated and currently interacting with an active "Service Info" tab environment. An employee's service data parameters have been loaded into active memory cache.
Post-conditions	The system validated that the NewBPS selection is within the legal administrative range of 1–22 and cross-checked that the NewDesignation adheres to legal court staff hierarchy mapping rules. Scenario A (Valid Update Details): - The corresponding attributes in the ServiceRecord class instance (Designation, BPS_Level, and Department) were modified in application memory. - A SQL UPDATE command transaction was successfully executed and committed to the database. - A system verification message notification ("Confirmation of professional status update") was generated and displayed to the Admin. Scenario B (Validation Error - Illegal Designation/BPS Mapping Conflict):

Field	Description
	- The local SQL database transaction was completely blocked and rolled back to preserve strict data integrity. - The application state retained its immediate pre-submission cache layout. - An on-screen error dialogue was rendered to prompt the Admin to select a valid, legally compatible judicial role.
Exceptions	InvalidDesignationBPMMappingException: Fired if business rule scripts detect an unmapped combination (e.g., matching a low-grade support role to a high BPS rank value).
Output	Commits an active update row block modification across the database storage system.
Notes	Uses transaction checks to ensure changes are written correctly even during unforeseen power fluctuations or application interruptions.

18.4 SSD4, UC04: Manage Postings & Transfers

18.4.1 Contract 7: Transfer Form Initialization Query

Field	Description
Operation	Display_Transfer_Form(PersonalNumber)

Field	Description
Type	MovementHistoryController / PostingTransfer
Cross References	UC04: Manage Postings & Transfers, FR-04: Posting & Transfer Log
Responsibility	Aggregates and visually maps the complete historical travel registry for an employee based on a unique identifier query.
Pre-conditions	The Authorized Admin is successfully logged into the system. A StaffProfile record matching the input PersonalNumber exists within the local database.
Post-conditions	The system executed a read-only lookup query using the provided identifier to fetch data from the database. The active data records for the employee's current station name, start date, and chronological posting logs were retrieved from the local SQL storage layer. The "Postings & History" graphical interface tab view was initialized and populated with these records for user visibility. The system state remained structurally unchanged with zero database writes executed.
Exceptions	EmptyHistoryException: Handled gracefully by the interface layer if there are no legacy transfers logged for a valid employee, displaying a clean initialization template.
Output	Array elements packed into an operational DataGridView list view collection block on screen.

Field	Description
Notes	Centralizing this tracking functionality significantly cuts down on structural manual errors associated with paper register management styles.

18.4.2 Contract 8: Posting and Transfer Transaction Execution

Field	Description
Operation	Execute_Transfer(NewStation, TransferDate, Reason)
Type	MovementHistoryController / PostingTransfer
Cross References	UC04: Manage Postings & Transfers, FR-04: Posting & Transfer Log
Responsibility	Concludes active historical registry logs and builds an updated structural transfer routing node inside a transaction block.
Pre-conditions	The Authorized Admin is authenticated and currently interacting with an active "Postings & History" view panel. The target employee's active posting record parameters have been successfully loaded into memory cache. Alphanumeric form parameters for NewStation, TransferDate, and Reason are fully provided.

Field	Description
Post-conditions	The system validated that the inputted TransferDate does not precede the staff member's historical Date of First Appointment and contains no timeline conflict relative to the active posting's start date. Scenario A (Valid Transfer & Timeline Details): - A local SQL database transaction block was opened to preserve relational consistency. - The active posting record for the employee was modified by setting its EndDate attribute to close out the assignment timeline. - A new PostingTransfer instance was instantiated with attributes set to the provided values (StationName = NewStation, StartDate = TransferDate, TransferReason = Reason) and linked to the corresponding StaffProfile entity. - The transaction successfully committed these writes across the database schema. - A success notification message box ("Transfer successful; Posting history updated") was generated and displayed to the Admin. Scenario B (Validation Fault - Chronological Timeline Conflict): - The database write transaction was completely blocked and rolled back to preserve data integrity. - The persistent data storage layer state was maintained in its immediate pre-submission layout. - An on-screen warning dialogue was triggered to prompt the Admin for a valid timeline correction.
Exceptions	ChronologicalTimelineException: Raised if incoming validation checks catch dates that violate sequencing guidelines.

Field	Description
Output	Structural changes committed atomically across the active database tables.
Notes	Relational design paradigms prevent deleting any profile data container if an active child log row is attached here.

18.5 SSD5, UC05: Manage Leave Records

18.5.1 Contract 9: Leave Balance Verification Query

Field	Description
Operation	Check_Leave_Entitlement(PersonalNumber)
Type	LeaveAllocationController / LeaveRecord
Cross References	UC05: Manage Leave Records, FR-05: Automated Leave Logic
Responsibility	Queries the system storage schema to display data blocks showing vacation metrics before parsing leaves.

Field	Description
Pre-conditions	The Authorized Admin is successfully authenticated into the system. A StaffProfile record matching the target input PersonalNumber exists within the local database layer.
Post-conditions	The system executed a read-only lookup query against the database using the provided PersonalNumber identifier. The active runtime attributes for the employee's DaysAvailed and RemainingBalance were successfully queried from the LeaveRecord database entity. The system initialized and populated the corresponding interface labels on the "Leave Management" module view screen to display these balances to the Admin. The persistent state of the system remained unchanged with zero database writes executed.
Exceptions	RecordNotFoundException: Triggered if data tracking scripts cannot establish matching primary key profiles.
Output	Updates variable interface strings on labels within the system frame.
Notes	This operation acts as a foundational safety mechanism preventing manual visual estimation errors when reviewing complex paper balances.

18.5.2 Contract 10: Leave Request Transaction Processing

Field	Description
Operation	Apply_Leave(RequestedDays, StartDate, EndDate)
Type	LeaveAllocationController / LeaveRecord
Cross References	UC05: Manage Leave Records, FR-05: Automated Leave Logic
Responsibility	Evaluates mathematical rules to modify remaining time elements within a secure system script framework.
Pre-conditions	The Authorized Admin is authenticated and actively interacting with the "Leave Management" form interface. The target employee's current RemainingBalance value has been successfully loaded into the application's local execution memory cache. Alphanumeric form parameters for RequestedDays, StartDate, and EndDate are fully provided by the user.
Post-conditions	The system executed an internal business logic evaluation check to verify if the input value RequestedDays $\leq$ the currently cached RemainingBalance. Scenario A (Valid Request - Sufficient Leave Balance Available): - An atomic local SQL transaction block was opened to enforce strict database consistency.

Field	Description
	- The persistent LeaveRecord.DaysAvailed attribute was incremented by adding the RequestedDays integer value. - The persistent LeaveRecord.RemainingBalance attribute was decremented by subtracting the RequestedDays integer value . - A new historical logging record was generated and inserted into the LeaveHistory data store table, linking the transaction fields (StartDate, EndDate, and duration metrics) to the corresponding profile . - The transaction safely committed these changes to the storage layer . - A confirmation dialog alert window ("Leave recorded successfully...") alongside an updated balance summary report view was generated and displayed to the Admin . Scenario B (Validation Fault - Insufficient Balance Left): - The database modification write actions were completely blocked, preserving data records in their original state . - The application state was safely rolled back to its immediate pre-submission cache layout. - An on-screen error dialogue box ("Insufficient Balance") was triggered on the user interface window to notify the Admin and deny execution.
Exceptions	InsufficientLeaveBalanceException: Raised if processing calculations cross below legal thresholds.
Output	Modifies state variables and updates lines within structural ledger tables.

Field	Description
Notes	Computations scale relative to a static legal limit constraint block of 25 annual tracking days.

18.6 SSD6, UC06: Manage Seniority Lists

18.6.1 Contract 11: For the Seniority List Generation Request

Field	Description
Operation	Request_Seniority_Calculation(Designation)
Cross References	UC06: Manage Seniority Lists
Pre-conditions	The Authorized Admin is successfully authenticated into the desktop application. Active StaffProfile instances linked to valid ServiceRecord entities exist in the local database layer for the specified administrative group. Mandatory ranking timeline data attributes (AppointmentDate, PromotionDate, MeritNumber, and DateOfBirth) are fully populated for the targeted staff category.
Post-conditions	The system executed a read-only filter query block to parse the database and isolate all active employees whose current position matches the input Designation parameter. The system invoked an automated multi-tier sorting engine in application memory, prioritizing tie-breaking variables

Field	Description
	strictly according to the Khyber Pakhtunkhwa (KP) Judicial Service Rules framework: Tier 1 (Primary Priority): Ordered records in ascending chronological sequence by the first date of employment entry (AppointmentDate). Tier 2 (Secondary Tie-Breaker): If identical initialization dates are encountered, records are sorted chronologically by the current grade advancement milestone (PromotionDate). Tier 3 (Tertiary Tie-Breaker): If promotion timelines match, the system ranks records using the unique assignment sequence code generated during initial recruitment (MeritNumber). Tier 4 (Quaternary Tie-Breaker): If a complete priority tie persists, the system evaluates the staff age baseline (DateOfBirth), sorting the oldest employee into the higher rank position. A sequential numerical position variable was computed and mapped onto each sorted record instance inside a temporary collection object. The finalized dataset array was dynamically bound and rendered on the user interface screen using a tabular layout (DataGrid) to present names, sequential rankings, and service histories to the Admin. The system database storage state remained static and unchanged, with zero persistent structural writes performed.

18.7 SSD7, UC07: Generate HR Reports

18.7.1 Contract 12: Report Type Selection Initialization

Field	Details
Name	Select_Report_Type(Profile, LeaveHistory, or Seniority)

Field	Details
Type	ReportingSubsystem / ReportController
Cross Reference	UC07: Generate HR Reports, FR-07: Report Generation
Responsibility	Captures the targeted configuration category input and dynamically adjusts the user interface to display the necessary filtering criteria textboxes/selectors.
Pre-conditions	The Authorized Admin is successfully logged into the system and interacting with the primary application reporting module panel.
Post-conditions	The target report category parameter value was captured and cached in the active application memory state. The system evaluated the selection type to determine the required search constraints. The graphical user interface dynamically updated its form fields to expose the specific filter prompt layout needed for execution (e.g., prompting for an individual employee's PersonalNumber search box or a category-wide position Designation dropdown menu selection).
Exceptions	None (purely cosmetic/UI state adaptation step).
Output	None.

Field	Details
Notes	Prepares the runtime application environment for parameter validation and compilation before the query layer execution.

18.7.2 Contract 13: Filter Criteria Provision & Compilation

Field	Details
Name	Provide_Filter_Criteria(SelectionValue)
Type	ReportingSubsystem / ReportController
Cross Reference	UC07: Generate HR Reports, FR-07: Report Generation
Responsibility	Executes multi-table relational SQL lookup operations matching input constraints to serialize and compile an on-screen document print preview layout template.
Pre-conditions	The Admin has chosen a valid report configuration category. The system is displaying an active query prompt state awaiting input parameters.
Post-conditions	The system validated that the supplied SelectionValue parameter satisfies basic syntax requirements.

Field	Details
	Scenario A (Target Record Matches Exist): The system opened a read-only data access layer to parse the database storage cache. A multi-table relational SQL JOIN command query was compiled and executed across the StaffProfile, ServiceRecord, LeaveRecord, and PostingTransfer structural tables to aggregate matching record lines. The retrieved dataset stream was serialized and mapped into a standardized, official District Court Orakzai reporting document template structure layout in the application memory cache. An explicit modal user interface view component (Print Preview window frame) was initialized and rendered on-screen, presenting the scannable, formatted document content overview to the Admin. Scenario B (No Records Found Parameter Error): The structural database lookup engine returned an empty result block due to an invalid identifier or non-existent parameter tracking search. The creation of a document preview layout block was bypassed to prevent rendering corrupted or empty templates. An explicit graphical error dialog panel message box notification ("No records found") was triggered on the active monitor interface window to notify the actor.
Exceptions	InvalidFilterFormatException: Raised if the provided text string violates required data types (e.g., entering alphabetic characters into a strictly numeric identifier text input field).
Output	Serialized template data layout stream initialized in active memory buffer cache.
Notes	Optimizes execution speeds by pulling targeted data clusters locally from the .NET runtime cache without forcing persistent database writes.

18.7.3 Contract 14: Document Output Spooler Dispatch

Field	Details
Name	Request_Print()
Type	ReportingSubsystem / ReportController
Cross Reference	UC07: Generate HR Reports, FR-07: Report Generation
Responsibility	Compiles and dispatches the actively cached, formatted reporting layout preview data stream to an external physical printing device or local digital PDF generator device handler.
Pre-conditions	A document report layout template has been successfully populated with database attributes and remains actively loaded within the active runtime Print Preview memory buffer stream.
Post-conditions	The formatted report data stream layout template was compiled and dispatched via an external Windows API subsystem framework call directed to the system's selected target output destination handler (e.g., standard physical Court Laser Printer spooler queue execution or local digital PDF document compiler write block).An interface alert dialog notification status popup confirming document output dispatch was generated and presented on the active desktop screen. The application state maintained its current display context with zero modifications or structural alterations executed against persistent database storage entities.

Field	Details
Exceptions	PrinterUnreachableException: Fired if communication handshakes with connected local physical hardware devices fail or local operating system print spoolers return an active queue error code.
Output	Raw binary print command stream directed to the operating system's active default peripheral queue or PDF compiler stream.
Notes	Wraps the compiled dataset into official court header templates before pushing the stream out of the system boundary.

18.8 SSD8, UC08: Generate HR Reports

18.8.1 Contract 15: Secure Session Termination

Field	Details
Name	Logout()
Type	AuthenticationController / SystemAdmin
Cross Reference	UC08: C, FR-01: User Authentication

Field	Details
Responsibility	Securely tears down active administrative access permissions, flushes volatile memory caches, and locks down application entry by resetting the interface to the unauthenticated login screen.
Pre-conditions	The SystemAdmin is currently successfully authenticated, and an active user session token instance exists within the application's runtime memory cache.
Post-conditions	The active session instance and authentication flags in application RAM were completely destroyed and cleared. The system state was safely reverted to an unauthenticated status. The current instance of the MainDashboard user interface window was closed and removed from execution focus. The primary LoginWindow view component was re-instantiated, initialized, and rendered on the host monitor, completely blocking access to all underlying judicial staff data clusters until valid credentials are supplied again. The persistent local SQL database state remained unchanged with zero modifications executed to stored tables.
Exceptions	SessionNotFoundException: Thrown if an unexpected memory fault or crash occurs where a teardown action is fired but no valid runtime session reference can be located in application memory.
Output	Security tracking status timestamp log entry sent to local file trace system.
Notes	Acts as an immediate localized memory sweep. Because the system runs as a standalone local desktop application, completely closing out

Field	Details
	and recreating the view states guarantees that data remains locked if the Superintendent leaves the workstation terminal unattended.

19. SEQUENCE DIAGRAMS

19.1 Sequence diagram 1 on Contract 1: Login Form Initialization

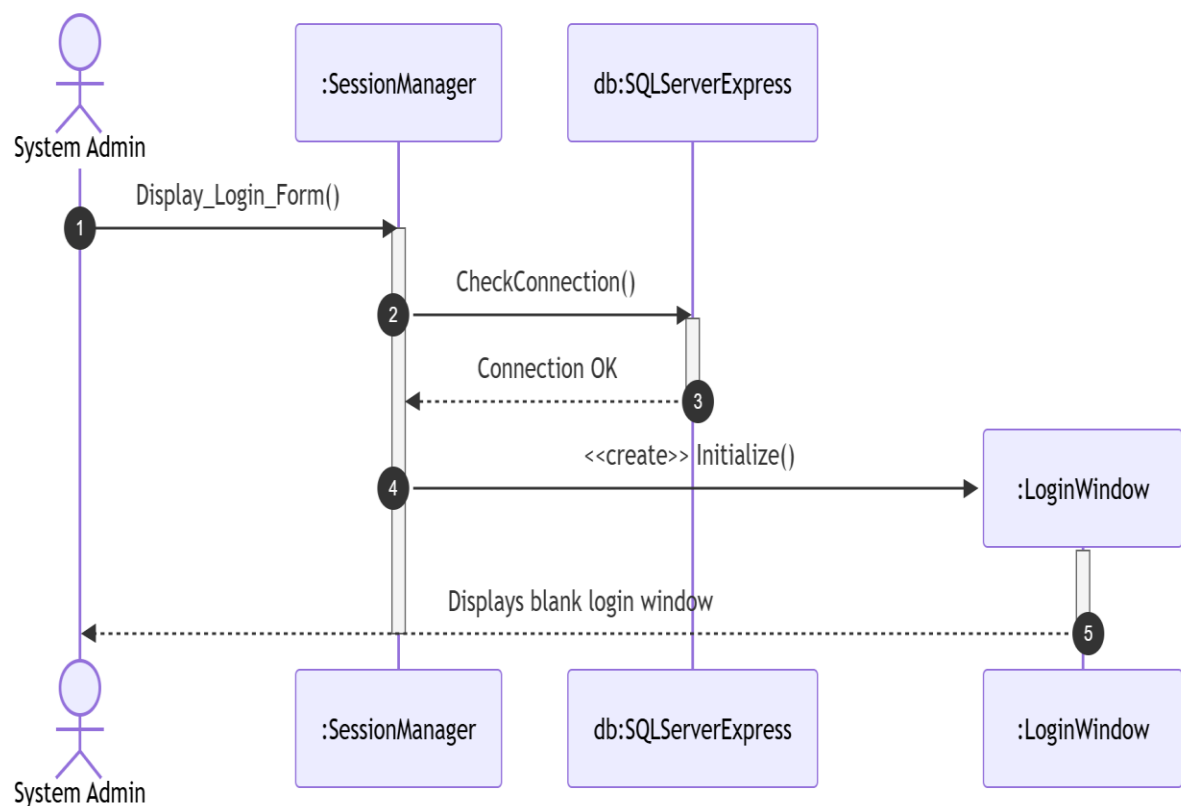


Figure 19.1: Sequence diagram for Contract 1 – `Display_Login_Form()` initializes the login security gateway.

19.2 Sequence Diagram 2 on Contract 2: User Identity Authentication

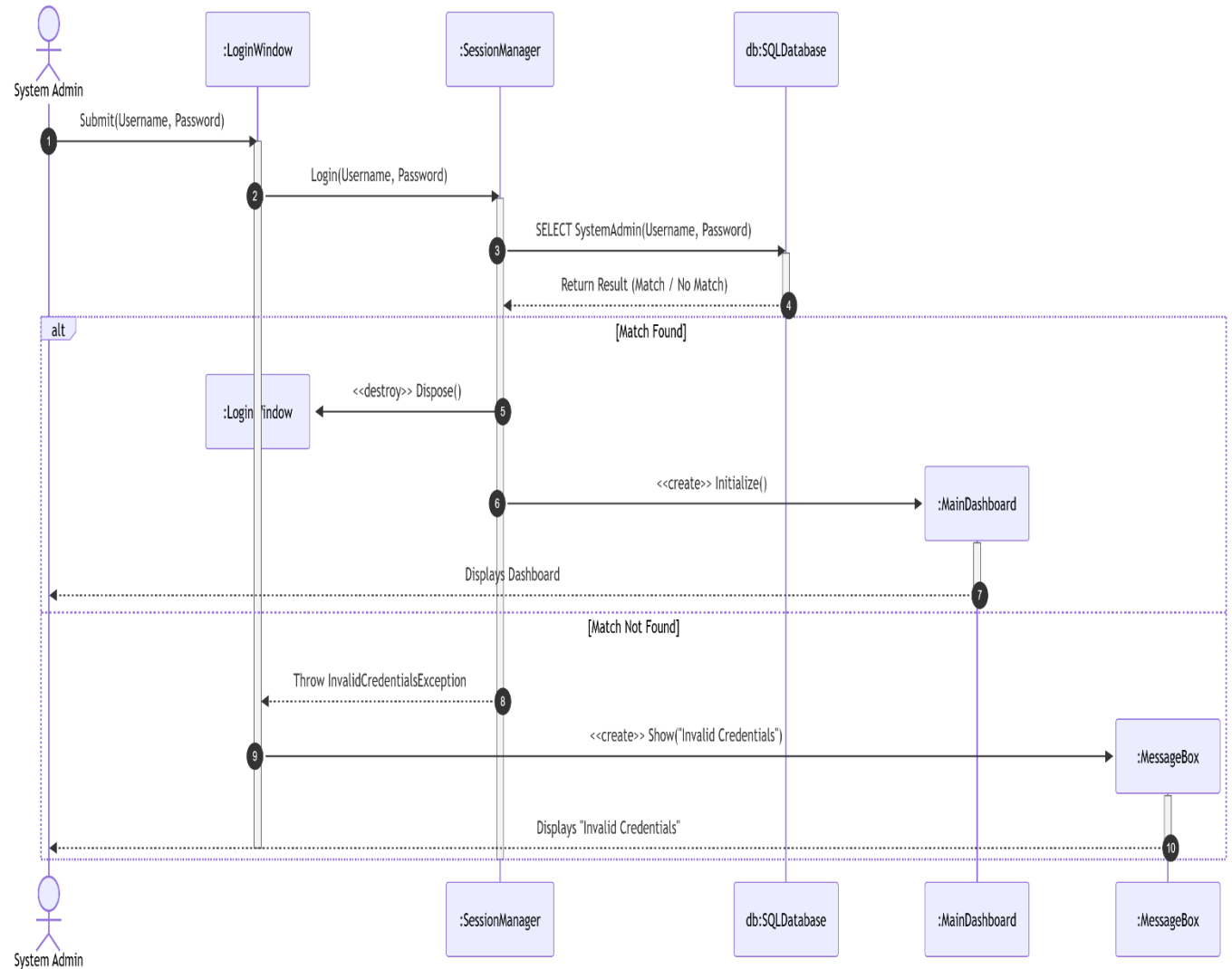


Figure 19.2: Sequence diagram for Contract 2 – Login() authenticates administrator credentials against the local database.

19.3 Sequence Diagram 3 on Contract 3: Profile Registration Form Initialization

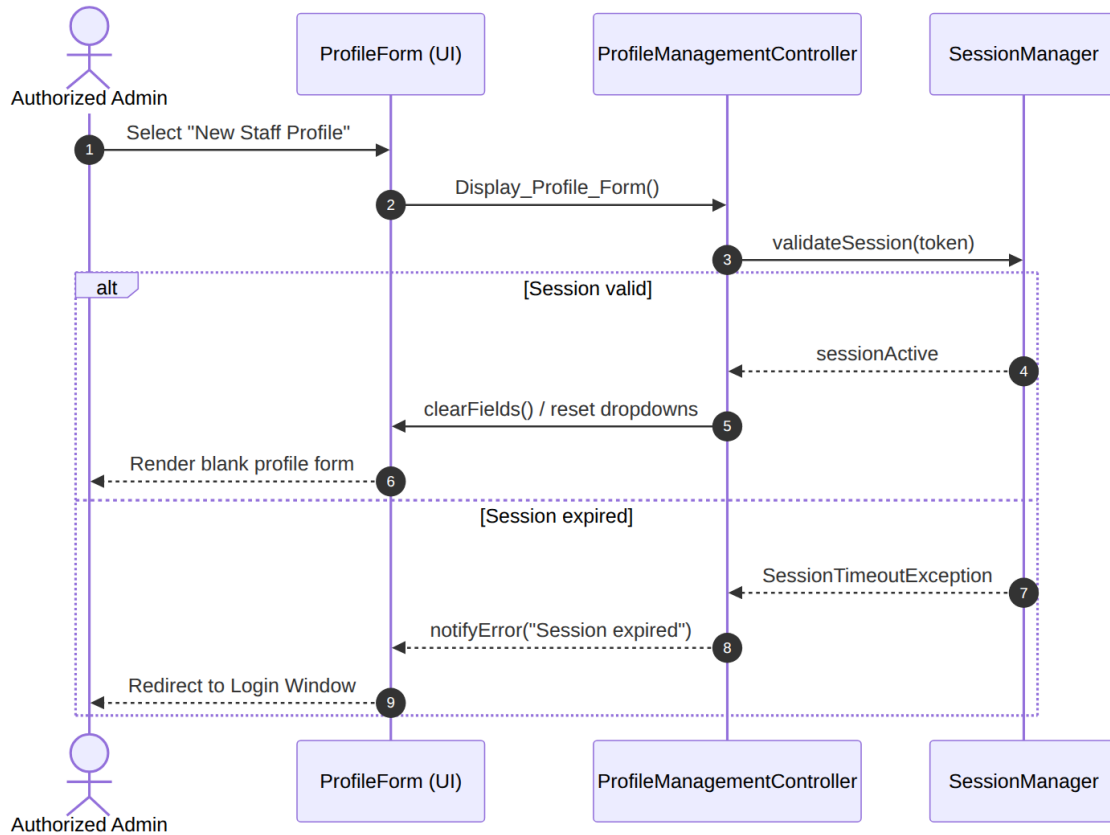


Figure 19.3: `Display_Profile_Form()` – `ProfileManagementController` initializes a blank profile entry form.

19.4 Sequence Diagram 4 on Contract 4: Profile Submission and Database Write

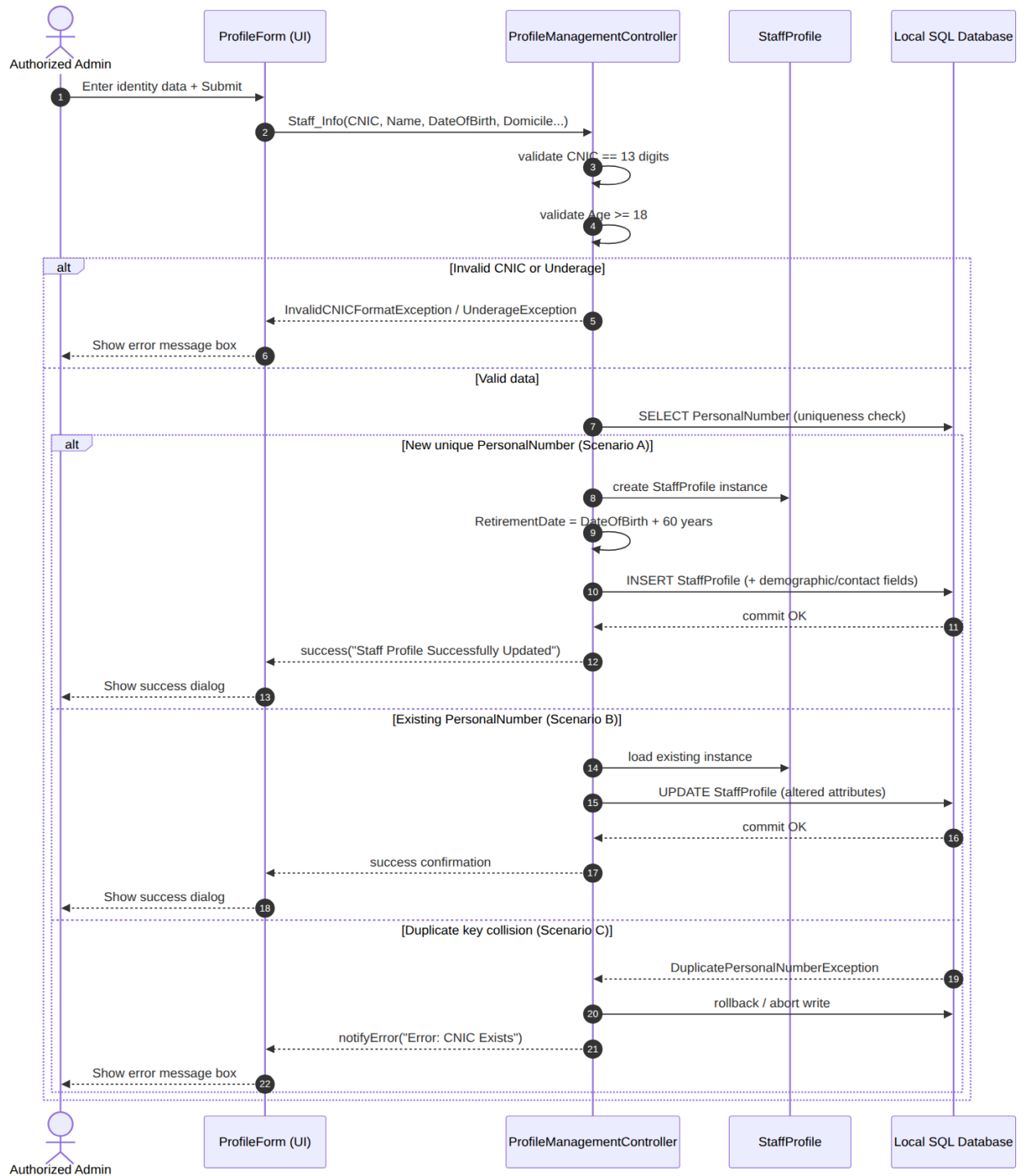


Figure 19.4: Staff_Info(...) – validation, INSERT/UPDATE and superannuation computation across Scenarios A–C.

19.5 Sequence Diagram 5 on Contract 5: Service Record Loading Query

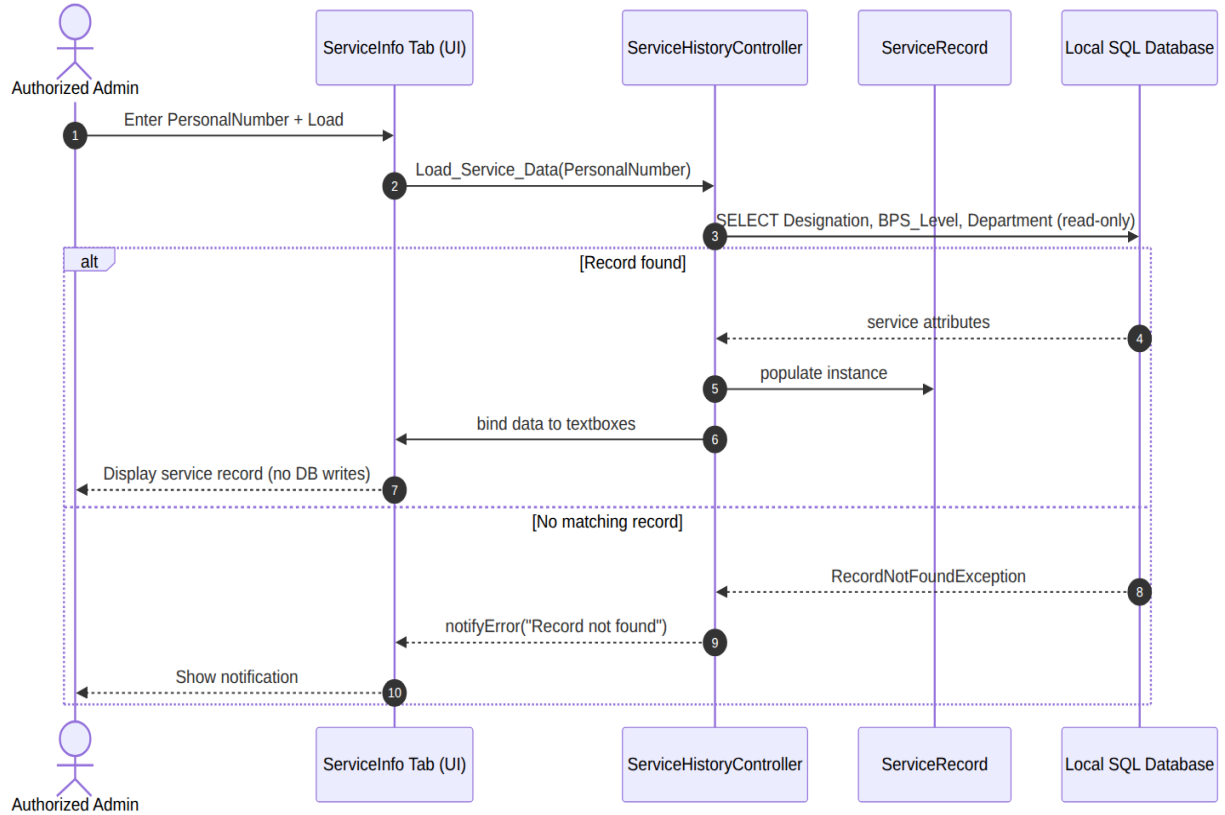


Figure 19.5: `Load_Service_Data(PersonalNumber)` – read-only retrieval of designation, BPS level and department.

19.6 Sequence Diagram 6 on Contract 6: Service Record Update Execution

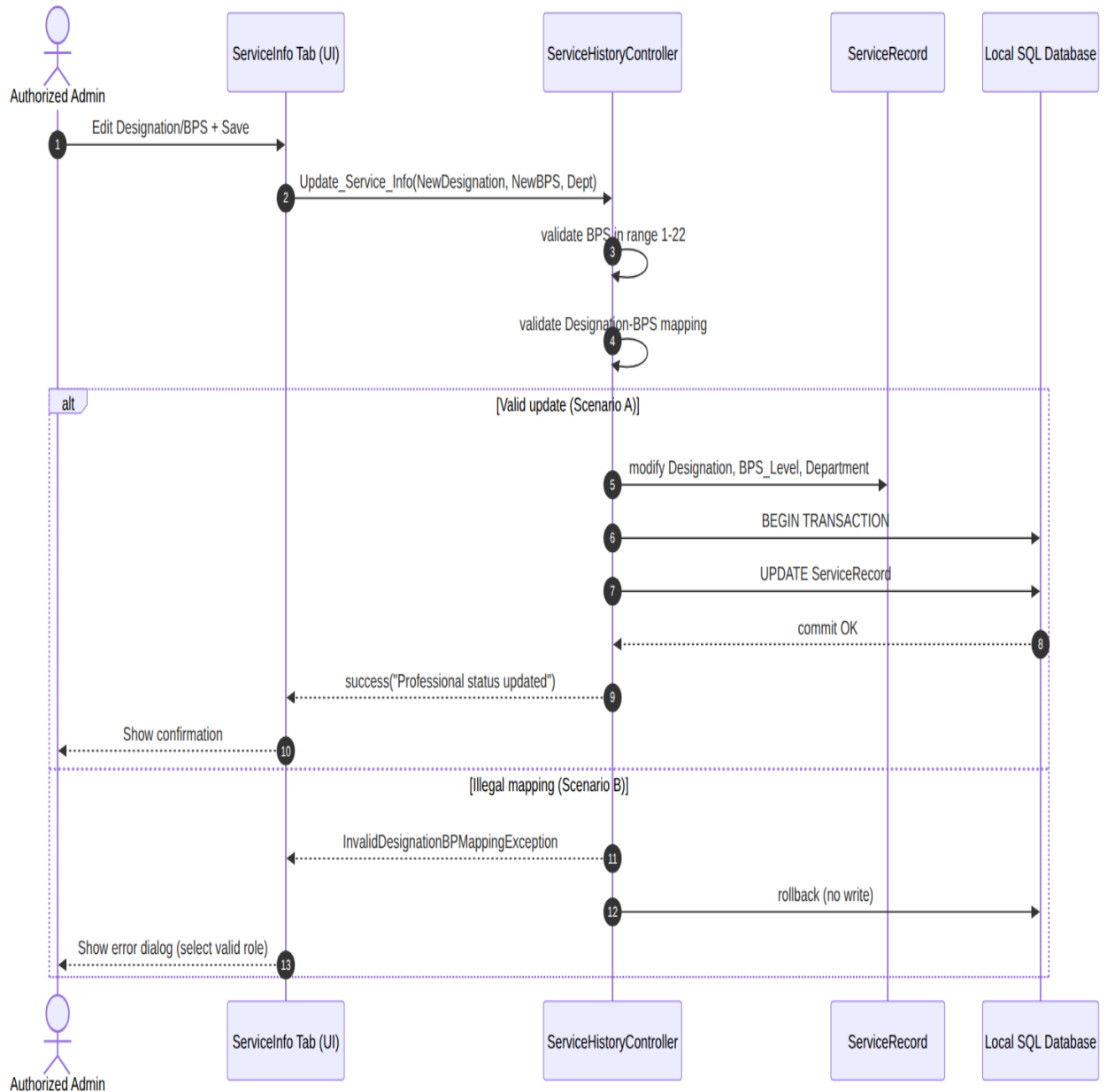


Figure 19.6: `Update_Service_Info(...)` – BPS range and Designation–BPS mapping validation before commit.

19.7 Sequence Diagram 7 on Contract 7: Transfer Form Initialization Query

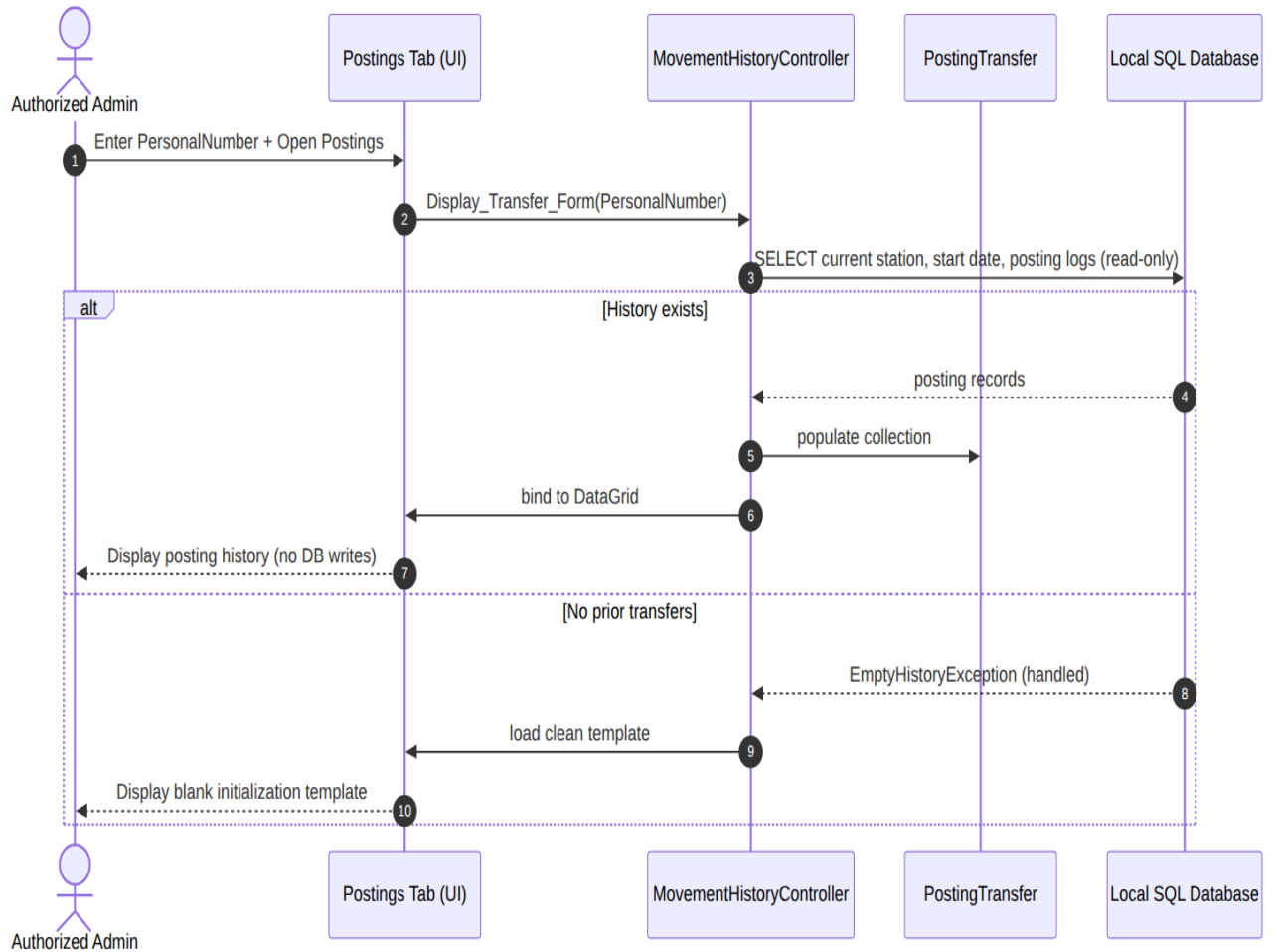


Figure 19.7: *Display_Transfer_Form(PersonalNumber)* – loads chronological posting history into the DataGridView.

19.8 Sequence Diagram 8 on Contract 8: Posting and Transfer Transaction Execution

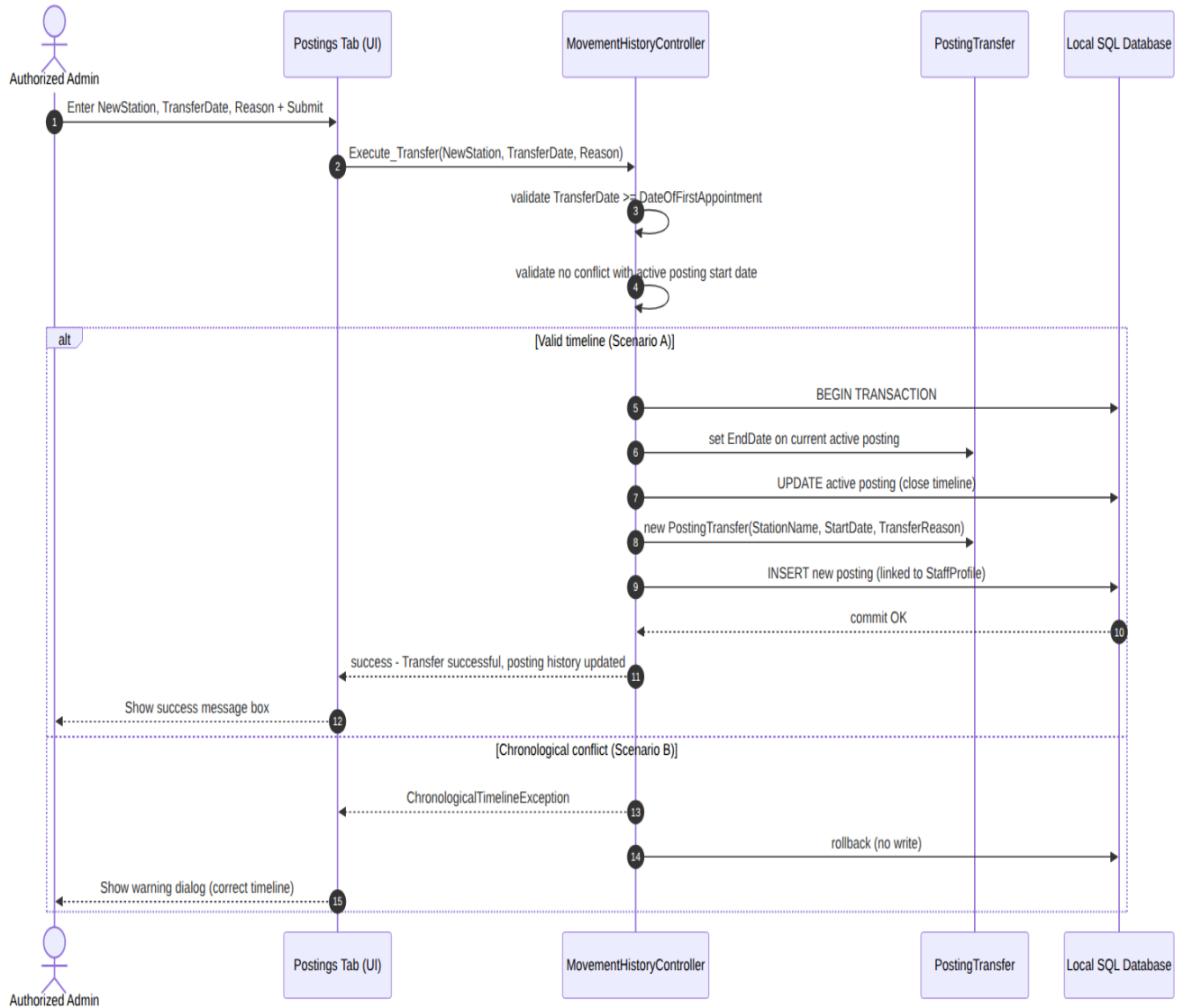


Figure 19.8: `Execute_Transfer(...)` – chronological validation, closing the active posting and inserting the new one.

19.9 Sequence Diagram 9 on Contract 9: Leave Balance Verification Query

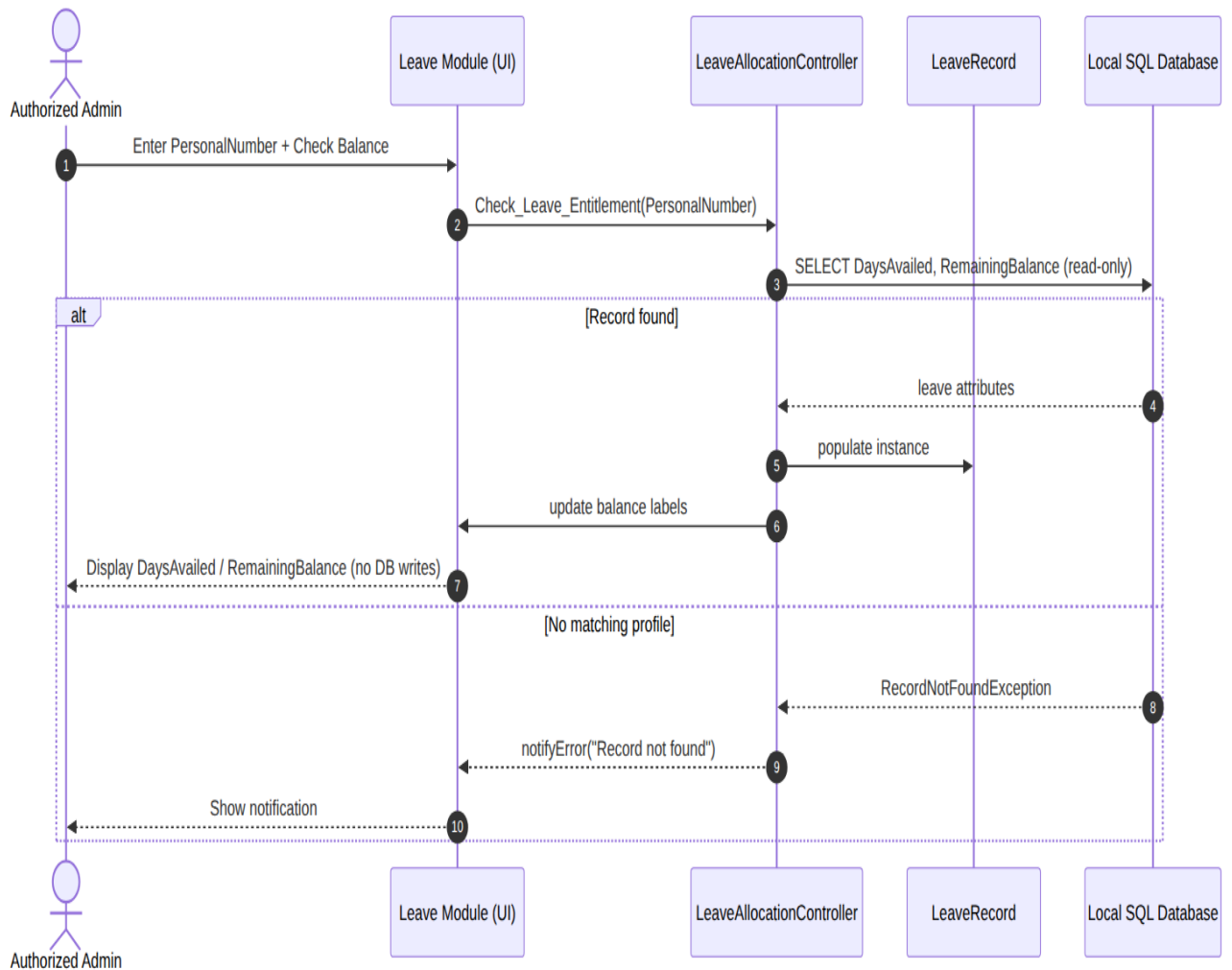


Figure 19.9: `Check_Leave_Entitlement(PersonalNumber)` – read-only display of availed and remaining leave.

19.10 Sequence Diagram 10 on Contract 10: Leave Request Transaction Processing

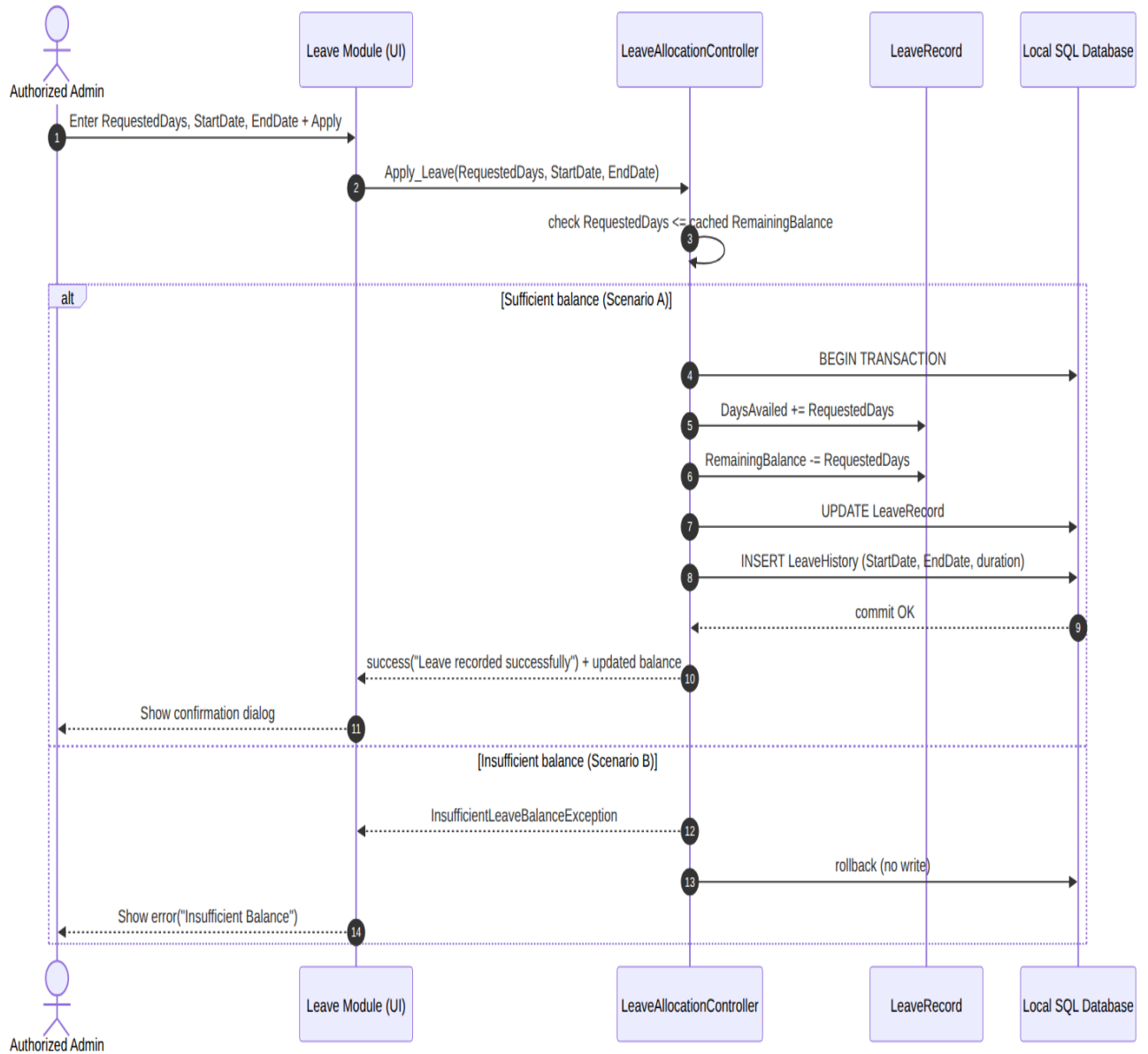


Figure 19.10: `Apply_Leave(...)` – balance check against the 25-day entitlement, atomic update and history logging.

19.11 Sequence Diagram 11 on Contract 11: Seniority List Generation Request

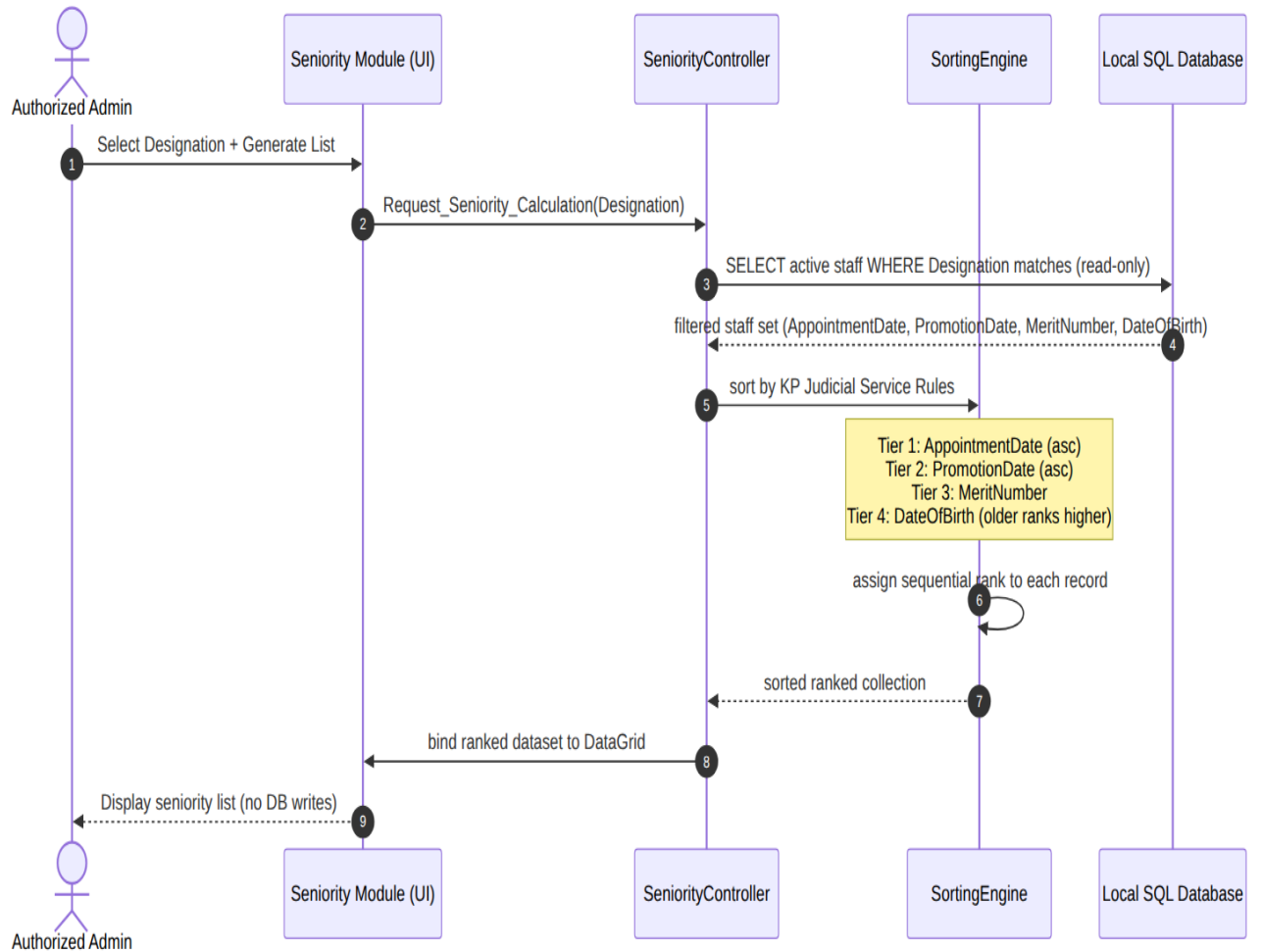


Figure 19.11: Request_Seniority_Calculation(Designation) – four-tier KP Judicial Service Rules ranking.

19.12 Sequence Diagram 12 on Contract 12: Report Type Selection Initialization

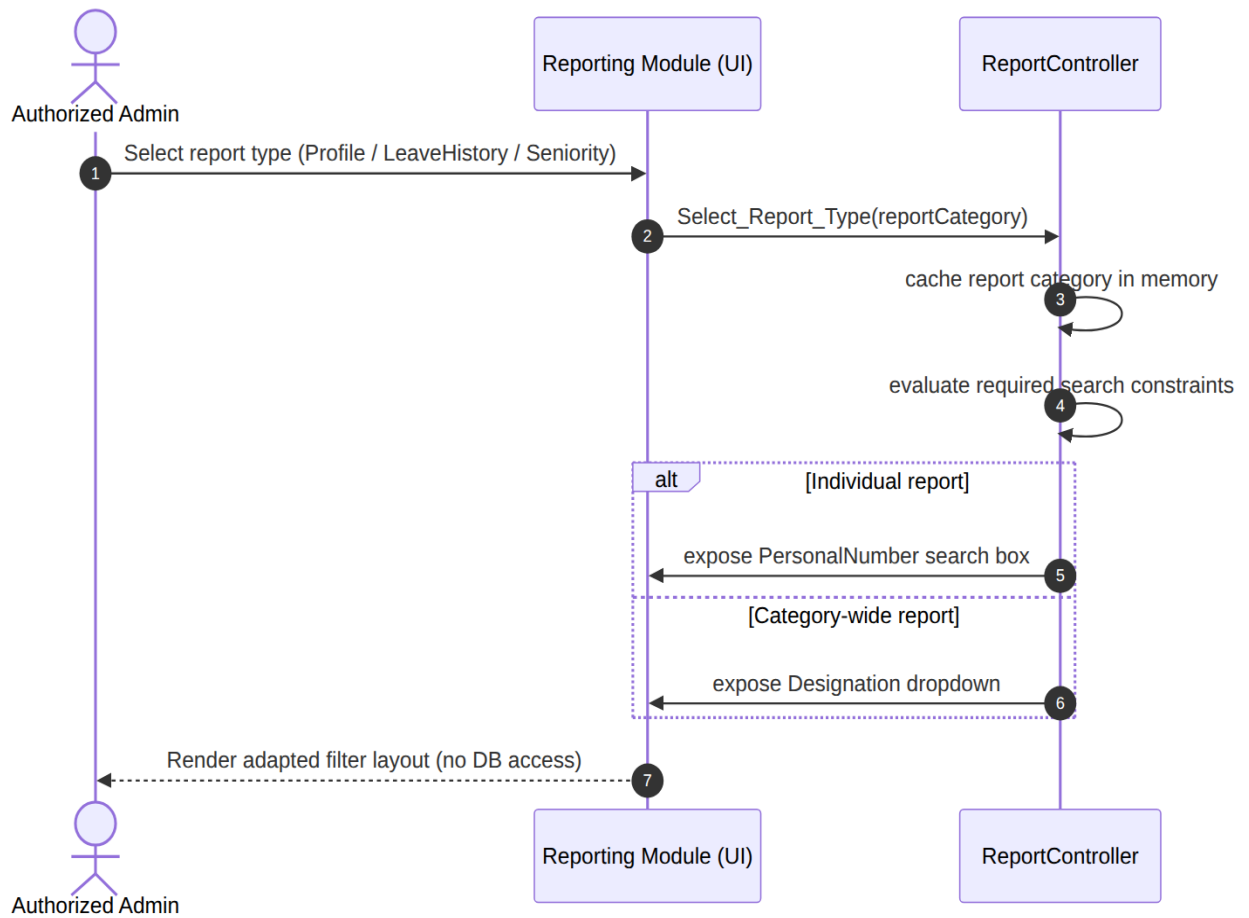


Figure 19.12: `Select_Report_Type(...)` – UI adapts filter fields to the chosen report category.

19.13 Sequence Diagram 13 on Contract 13: Filter Criteria Provision & Compilation

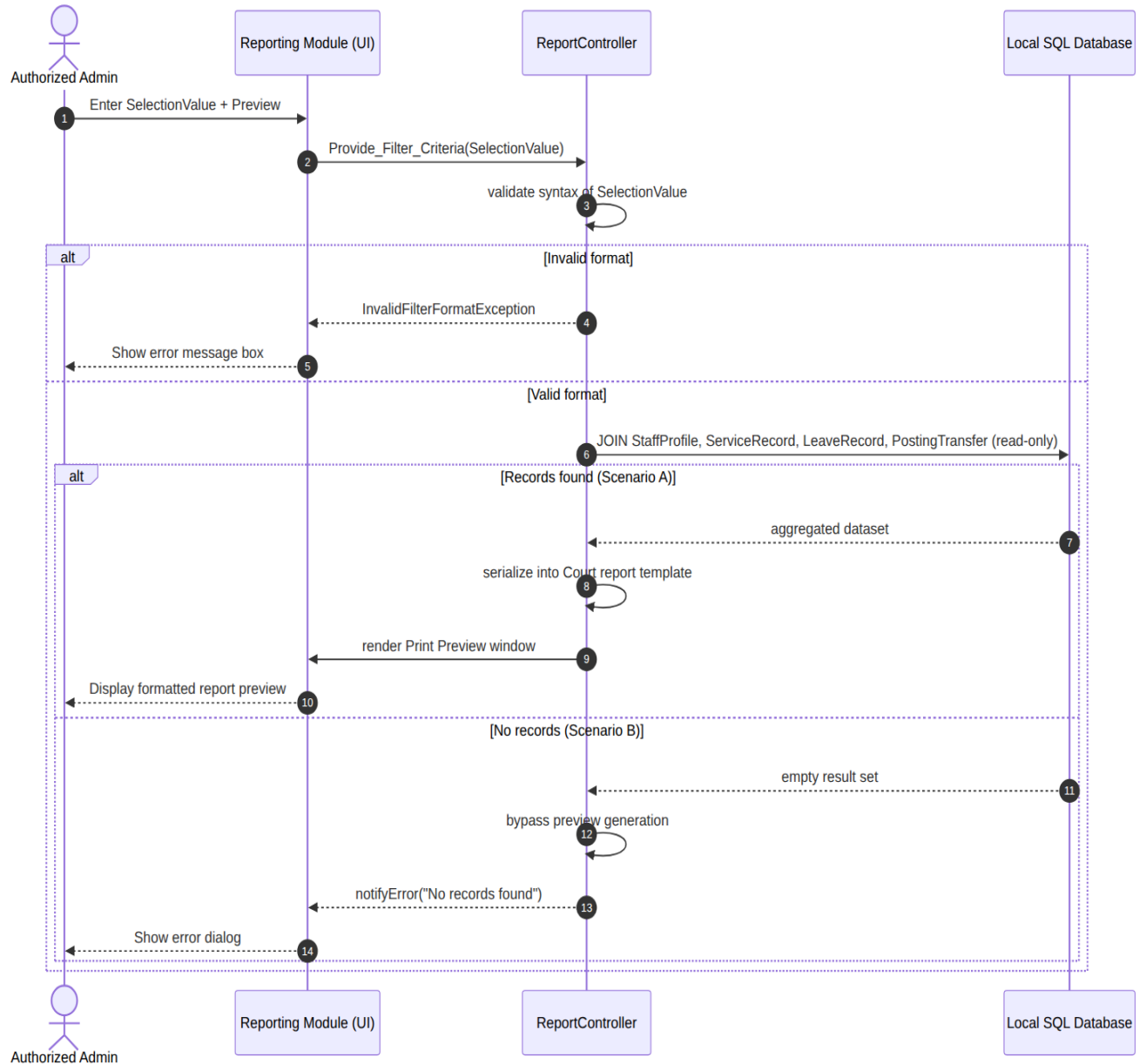


Figure 19.13: Provide_Filter_Criteria(SelectionValue) – multi-table JOIN and print-preview compilation.

19.14 Sequence Diagram 14 on Contract 14: Document Output Spooler Dispatch

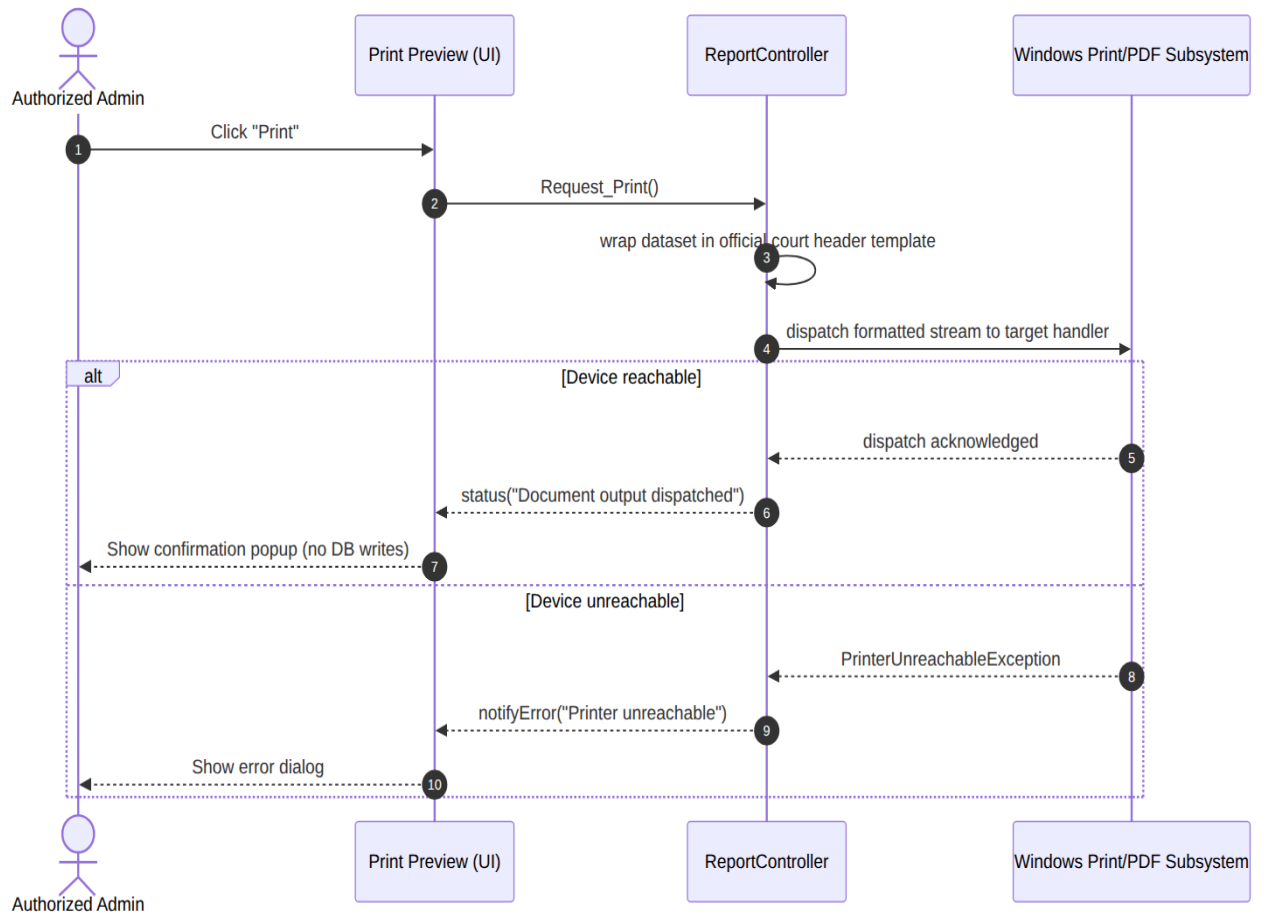


Figure 19.14: `Request_Print()` – dispatch of the formatted report stream to printer or PDF handler.

19.15 Sequence Diagram 15 on Contract 15: Secure Session Termination

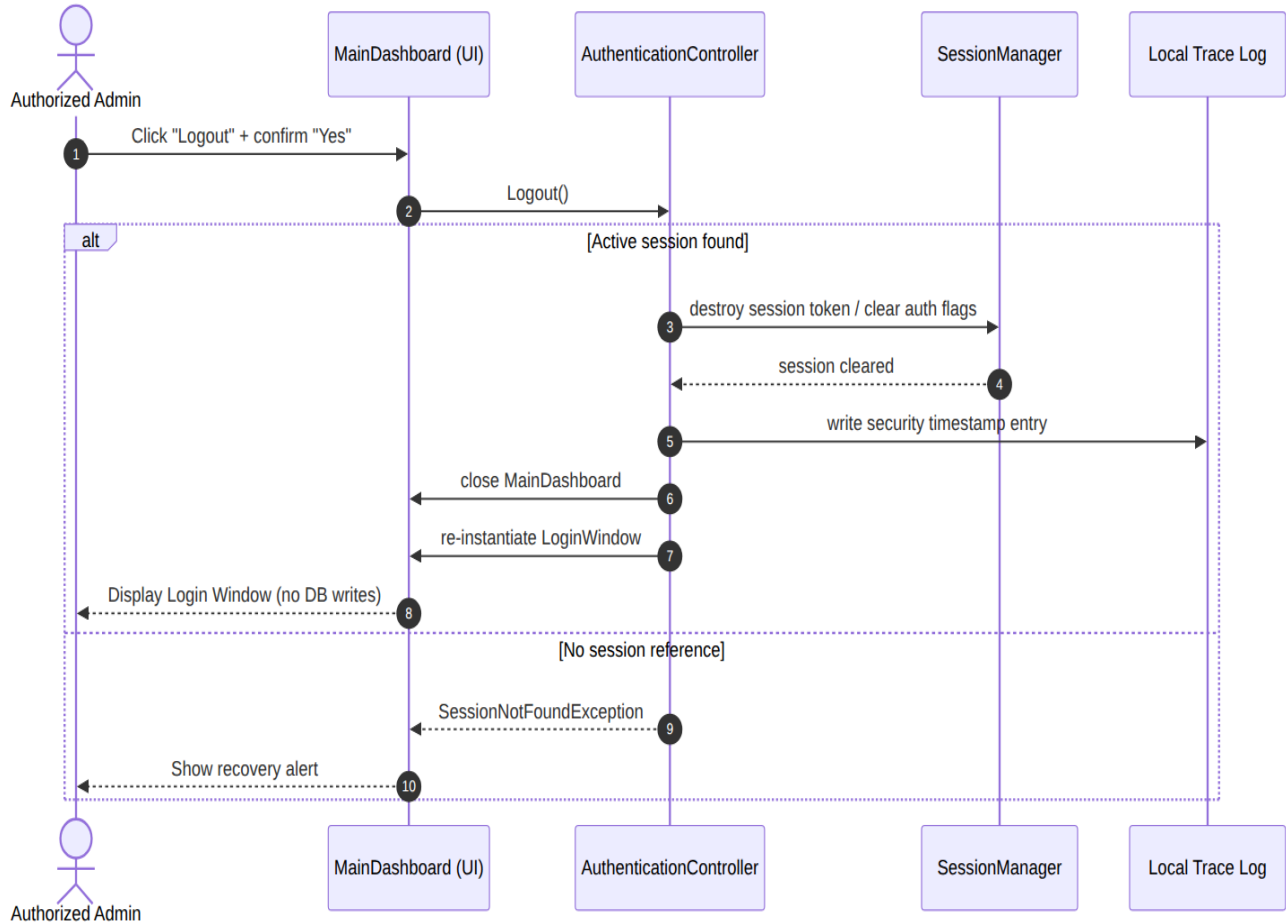


Figure 19.15: Logout() – session teardown, trace logging and return to the Login Window.

20. Package Diagram

The package diagram presents the high-level architectural organization of the HR Management System for District Court Orakzai. The standalone Windows desktop application is structured into five logical layers, enforcing a clear separation of concerns and ensuring that the sensitive judicial personnel data flows through controlled, well-defined boundaries.

Presentation Layer (UI): Contains all Windows GUI components the Superintendent interacts with – the Login Window, Main Dashboard, and the dedicated forms for profiles, service records, postings, leave, seniority, and reporting.

Application / Controller Layer: Houses the controllers that enforce business rules (e.g. the 25-day leave limit, BPS–Designation mapping, chronological transfer

validation, and KP Judicial Service Rules ranking). The UI never accesses data directly; all requests pass through this layer.

Domain / Entity Layer: Defines the core business objects – StaffProfile, ServiceRecord, PostingTransfer, LeaveRecord, LeaveHistory, and SystemAdmin – which model the real-world judicial HR concepts.

Data Access Layer: Encapsulates SQL data access and ACID-compliant transaction management, isolating the rest of the system from persistence concerns and protecting against data corruption during power fluctuations.

Infrastructure / OS Services: Represents external resources the system depends on – the local SQL Server Express database, the Windows print/PDF subsystem, and the local trace log for security auditing.

20.1 System Architecture

The HR Management System for District Court Orakzai adopts a Layered (N-tier) architecture organized along Model–View–Controller (MVC) lines, deployed as a single standalone (monolithic) Windows desktop executable. The system is divided into five logical layers – Presentation (UI), Application/Controller, Domain/Entity, Data Access, and Infrastructure – in which each layer communicates only with the layer directly beneath it. The user interface never accesses the database directly; every request is mediated by a dedicated controller that enforces the relevant business rules before any persistent operation is performed.

This style was selected for the following reasons:

- **Separation of Concerns:** Isolating the UI, business logic, and data access into distinct layers keeps the sensitive judicial personnel data behind controlled boundaries and makes the seniority, leave, and validation logic easier to verify against the Khyber Pakhtunkhwa Judicial Service Rules.
- **Alignment with the V-Model:** The disciplined, well-defined boundaries of a layered design map cleanly onto the stage-wise verification of the chosen V-Model process, allowing each layer (e.g. the Leave module's controller) to be unit- and integration-tested against its specification.
- **Maintainability and Low Coupling:** Because dependencies flow in one direction only, a change to the user interface or the report format does not ripple into the domain or data layers, reducing the risk of introducing regressions during maintenance.

- **Data Integrity:** Centralizing all persistence through a single Data Access layer with ACID-compliant transaction management ensures that leave balances and posting histories cannot be corrupted by partial writes during power fluctuations.
- **Suitability of Monolithic Deployment:** As an offline, single-user tool intended to run on existing court hardware, a self-contained desktop application avoids the network dependencies, operational overhead, and security exposure that distributed styles such as microservices would introduce.

In summary, the project is best described as a monolithic, standalone desktop application built on a layered (N-tier) architecture with an MVC-style separation of views, controllers, and domain entities, backed by a local relational data store. The package diagram below illustrates this layered organization and the permitted dependencies between the layers.

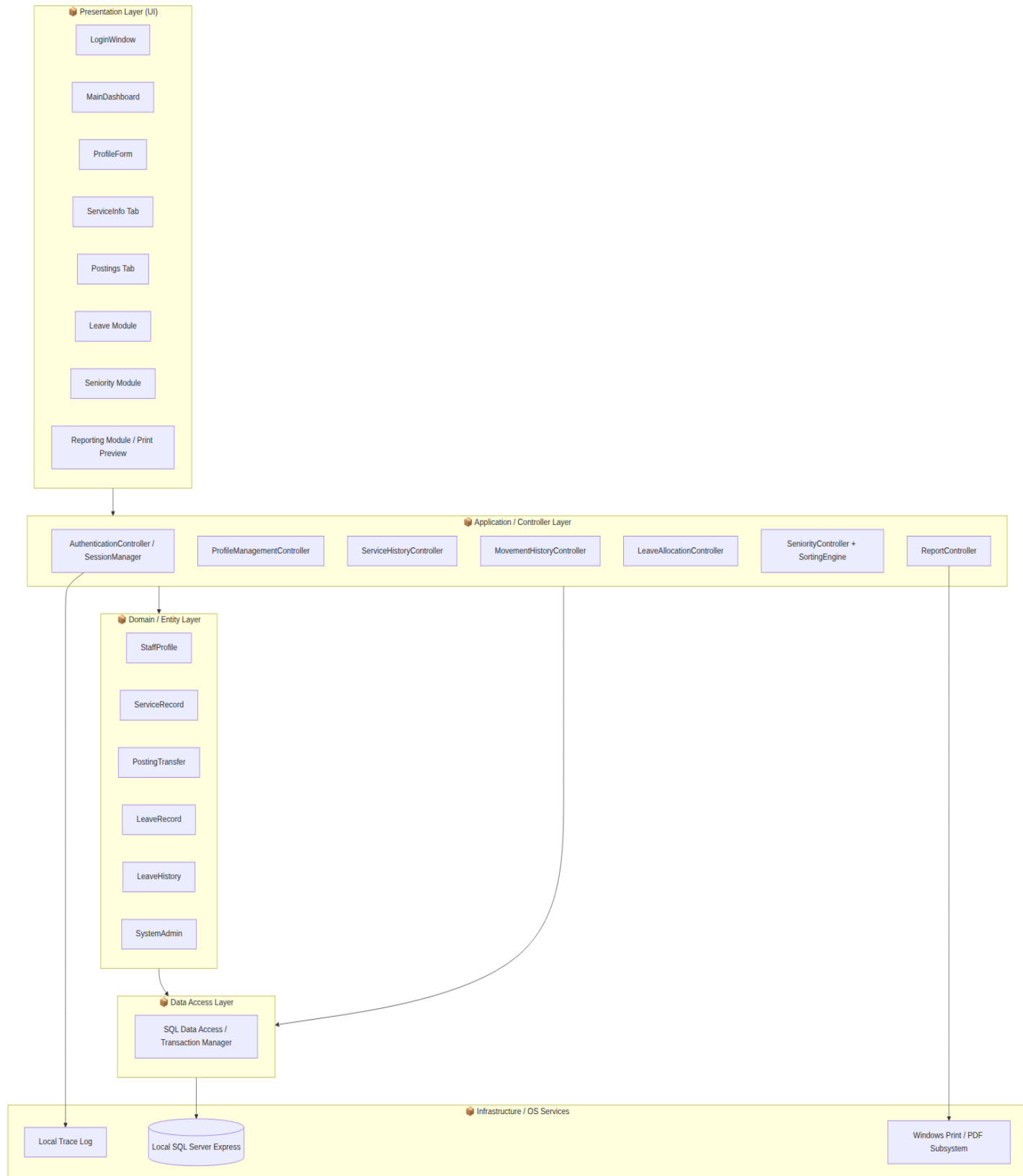


Figure 20.1: Layered package architecture of the District Court Orakzai HR Management System.

21. Definitions, Acronyms & Abbreviations

This section defines the domain-specific terms, acronyms, and abbreviations used throughout this document, to ensure that the judicial administration, developers, and any future maintainers share a common vocabulary.

Term / Acronym	Definition
AD&SJ	Additional District & Sessions Judge – a high-level judicial officer ranking below the District & Sessions Judge.
BPS	Basic Pay Scale – the government pay-grade system (levels 1–22) used to classify the rank of court employees.
CNIC	Computerized National Identity Card – the 13-digit national identifier issued to Pakistani citizens.
D&SJ	District & Sessions Judge – the final authority over HR and administrative decisions at the court.
DFD	Data Flow Diagram – a graphical representation of how data moves through the system.
Domicile	The district/province of permanent residence recorded for a staff member, used for quota and administrative purposes.
GUI	Graphical User Interface – the visual, window-based interface of the desktop application.
HR	Human Resource – the management of staff records, service history, leave, and seniority.
KP	Khyber Pakhtunkhwa – the province in which District Court Orakzai is located.
KP Judicial Service Rules	The provincial legal framework governing seniority, leave, and service conditions of judicial staff.
Naib Qasid	A support-staff designation (typically BPS-1 to BPS-4) responsible for office assistance and errands.
Personal Number	A unique identifier assigned to each staff member, used as the primary key for their records.
RBAC	Role-Based Access Control – tiered permissions; explicitly excluded from this single-user system's scope.

Seniority	The ranking of staff within a designation, computed from appointment and promotion dates per KP rules.
Service Book	The traditional paper register recording an employee's professional lifecycle, which this system digitizes.
SRS	Software Requirements Specification – this document.
SSD	System Sequence Diagram – a diagram showing the events between an actor and the system for a use case.
Steno	Stenographer – an administrative staff designation responsible for dictation and transcription.
Superannuation	Mandatory retirement; the system computes the retirement date as Date of Birth + 60 years.
Superintendent	The senior administrative officer who acts as the Authorized User/Admin of this system.
UC	Use Case – a description of a system–actor interaction achieving a goal.

22. User Characteristics

The system is designed for a single class of operator within the court's administrative wing. Understanding the characteristics of this user informs the usability and training requirements of the application.

22.1 Primary User: The Authorized Administrator

The primary (and only) operator of the system is the Superintendent or Clerk of Court, acting as the Authorized Administrator. This user is assumed to have the following characteristics:

Characteristic	Description
Role	Superintendent / Clerk of Court, operating under the supervision of the District & Sessions Judge.
Technical Skill	Baseline familiarity with the Windows operating system and standard GUI navigation (forms, buttons, dropdowns, dialogs); not a technical or programming background.

Domain Knowledge	Strong working knowledge of court HR procedures – service records, postings, the 25-day leave entitlement, and KP seniority rules.
Frequency of Use	Regular, day-to-day use for data entry, record updates, leave processing, and report generation.
Access Level	Full administrative rights. As tiered access (RBAC) is out of scope, every authenticated user has identical, complete privileges.
Training Need	Minimal. The interface uses standard Windows conventions so that little or no formal training is required.

22.2 Indirect Stakeholder: The District & Sessions Judge

While the District & Sessions Judge does not operate the application directly, the Judge is the approving authority for the final printed reports (seniority lists, individual profiles, and service histories). The Judge is therefore an indirect stakeholder whose needs shape the reporting outputs but who does not interact with the data-entry interface.

23. Use Case Ranking

Following the Larman method, the use cases are ranked by their significance to the architecture and the project's risk profile. This ranking guided the order in which use cases were analyzed and designed, ensuring that the most architecturally significant and risky functionality was addressed first. The ranking criteria are: (a) significant impact on the architecture or database design, (b) significant information or reporting value to the client, and (c) risky, time-critical, or algorithmically complex logic.

Rank	Use Case	Ranking Justification (Impact / Value / Risk)
1	UC06: Manage Seniority Lists	High architectural impact and high complexity – the four-tier KP ranking algorithm is the most logic-intensive and risk-prone feature.
2	UC05: Manage Leave Records	High value and moderate risk – enforces the mandatory 25-day entitlement with atomic balance updates that must survive power failures.

3	UC02: Manage Staff Profiles	High architectural impact – the StaffProfile entity is the foundational record on which all other modules depend.
4	UC01: Login to the System	Architecturally significant – the security gateway that all other use cases depend upon.
5	UC04: Manage Postings & Transfers	Moderate value and risk – chronological validation and the transactional close-and-open of posting records.
6	UC03: Manage Service Records	Moderate value – maintains designation/BPS data with business-rule validation.
7	UC07: Generate HR Reports	High information value – produces the printed outputs reviewed and approved by the District & Sessions Judge.
8	UC08: Logout from System	Low impact – a supporting use case that securely terminates the session.